Multiple-photon transitions in EPR spectroscopy

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Abstract

An overview of the various multiple-photon processes in electron paramagnetic resonance (EPR) spectroscopy is given. First, we describe different types of multiple-photon transitions that can be observed with monochromatic and bichromatic microwave fields in spin systems with unequally spaced energy levels and in two-level systems. Then we discuss multiple-photon processes that are based on a bichromatic radiation field consisting of a transverse microwave field and a longitudinal radio frequency field. Two semiclassical methods, namely the tilted frame and the toggling frame approach, as well as the quantized radiation field formalism (second quantization), and Floquet theory are used for the theoretical description of the effects. We discuss how these processes manifest in the conventional field modulated continuous wave EPR experiment and demonstrate how multiple-photon transition can be observed in pulse EPR. Finally, new features of multiple-photon transitions induced by bichromatic radiation fields are introduced. We particularly stress on the phenomenon of π-photon-induced transparency and describe the characteristics of these transitions and their potential applications.

1 Introduction

Many phenomena in infrared spectroscopy and coherent optics are based on the interaction of electromagnetic radiation with matter, where several photons are simultaneously absorbed or emitted [1–7]. Such multiple-photon transitions[1], have already been described in the first half of the last century [8]. Also in magnetic resonance multiple-photon phenomena have been known for a long time, but up to now extensive use of them as a spectroscopic tool has not yet been made.

In any transition between two energy levels energy and angular momentum are conserved. As a consequence, during a multiple-photon process in magnetic resonance the energy and the polarization of the photons involved in the process must correspond to the total change in energy and angular momentum in the spin system.

Electrons have an intrinsic angular momentum characterized by the spin quantum number $S = 1/2$. When an electron spin is subjected to a static magnetic field $\mathbf{B}_0$ oriented along the laboratory z-axis, the projection of the angular momentum along this axis is either $\hbar/2$ or -$\hbar/2$, and the energy levels corresponding to the magnetic quantum numbers $m_S = 1/2$ and $m_S = -1/2$, are split owing to the Zeeman interaction.

Photons have an intrinsic angular momentum $\hbar$, characterized by the spin quantum number $J = 1$. Photons in state $m_J = -1$ have angular momentum -$\hbar$ relative to the direction of propagation and are called σ^- photons (negative helicity), whereas photons in state $m_J = 1$ have angular momentum $\hbar$ and are called σ^+ photons (positive helicity) [3]. Left and right circular fields perpendicular to $\mathbf{B}_0$ can be associated with σ^- and σ^+ photons propagating along $\mathbf{B}_0$.

The state with projection $m_J = 0$ is forbidden, since a photon has a vanishing mass. However, a linear field oriented parallel to $\mathbf{B}_0$ can be associated with so-called π photons propagating perpendicular to $\mathbf{B}_0$ and described by the wave function $\pi = (\sqrt{2}/2)\sigma^+ + (\sqrt{2}/2)\sigma^-$. The eigenstate of this function thus corresponds to $m_J = 0$, and absorption or emission of π photons does not change the total angular momentum of the spin system.

In magnetic resonance radio frequency (rf) or microwave (mw) radiation fields are applied. A radiation field with only one frequency is called a *monochromatic* field. Correspondingly, we call a radiation field with two frequencies a *bichromatic* field, independent on whether the frequencies are very close to each other or differ by orders of magnitudes.

During a multiple-photon process, several photons are simultaneously absorbed and/or emitted from the radiation field. The sum of the angular momenta of all absorbed σ photons must equal the change in angular momentum of the electron spin system expressed by the change in m_S. There is no restriction to the number of absorbed or emitted π photons, since they do not add to the angular

[1] In literature the term "multiple-quantum transition" is often used instead of "multiple-photon transition". However, since in magnetic resonance this term is also used for transitions between states which differ in the magnetic quantum number by $\Delta m > 1$, we use the term multiple-photon transition throughout this work. During a multiple-quantum transition the radiation frequency matches the energy difference between the two levels involved in the transition so that, like in a single-quantum transition, only one photon is absorbed or emitted. On the other hand, during a multiple-photon transition several photons are absorbed and/or emitted.

momentum.

Several theoretical approaches have been developed to describe multiple-photon processes [9]. In magnetic resonance the Bloch equations [10, 11], or semi-classical theories, with the radiation field treated as a classical observable, are frequently used. However, to understand the physics behind a multiple-photon process second quantization, where not only the spins but also the radiation fields are quantized [12–14], and Floquet theory [15–17] has to be applied. A brief introduction to second quantization and Floquet theory is given in the Appendix.

In this Lecture Notes we discuss different types of multiple-photon processes that can be observed in electron paramagnetic resonance (EPR) spectroscopy. Completeness in the cited literature is not aimed at. In *pulse* EPR spectroscopy multiple-photon resonances are also attractive from the point of view of spin dynamics and potential applications. In Section 2 we give a brief overview of the different types of multiple-photon transitions that can be observed in EPR with monochromatic or bichromatic mw fields. Section 3 is devoted to multiple-photon transitions induced by a bichromatic radiation field consisting of a transverse mw field and a longitudinal rf field. In Section 3.1 we describe how these processes manifest in continuous wave (cw) EPR. Although this type of multiple-photon transitions is omnipresent in cw EPR, most spectroscopists are not aware of the fact that they are dealing with them in their daily routine work. In Section 3.2 we demonstrate how multiple-photon transitions can be observed in pulse EPR. Finally, in Section 3.3 a new type of multiple-photon transitions induced by a bichromatic field is introduced. We particularly stress on the phenomenon of π-photon-induced transparency and describe the characteristics of these transitions and their potential applications.

2 Different types of multiple-photon transitions in EPR

The energies of states of a paramagnetic species with an electron spin S and nuclei with spins I subject to a static magnetic field $\mathbf{B}_0$ are described by the static spin Hamiltonian $\mathcal{H}_0$. The Hamiltonian consists of terms which describe the interactions between the electron spin and $\mathbf{B}_0$, as well as the interaction between electron spins and between electron and nuclear spins.

The coupling between an electron spin and a linear radiation field $B_{\rm mw}(t) = 2B_1\cos(\omega_{\rm mw}t)$ with amplitude $2B_1$ and mw frequency $\omega_{\rm mw}$ is expressed by the perturbation Hamiltonian $\mathcal{H}_1(t)$ [18]. In this Lecture Notes energies and amplitudes of radiation fields are usually given in angular frequency units, i.e., $\epsilon = E/\hbar$ and $\omega_1 = -\gamma_e B_1$.

In the great majority of EPR studies, the linear response of a spin system to a weak radiation field is recorded. The transition amplitudes are then determined by the matrix elements of the perturbation operator $\mathcal{H}_1(t)$. The transition amplitude for a *single-photon transition* between the two eigenstates with energies ϵ_1 (labeled by 1) and ϵ_2 (labeled by 2) of the static Hamiltonian having wave functions ϕ_1 and ϕ_2 is given by

$$a_{12} = < \phi_1 \mid \mathcal{H}_1 \mid \phi_2 > , \tag{1}$$

where $\mathcal{H}_1$ is the *time-independent* perturbation Hamiltonian in a frame that rotates with the mw frequency ω_{mw} (rotating frame). The computation of a_{12} is most convenient in the eigenbasis of $\mathcal{H}_0$, where a_{12} is the matrix element of $\mathcal{H}_1$ connecting the states 1 and 2.

Due to state mixing, the magnetic quantum numbers of the electron and nuclear spins may no longer be good quantum numbers, and forbidden EPR transitions of the type $(\Delta m_S = \pm 1; \Delta m_I = \pm 1, \pm 2, ...)$, or $(\Delta m_S = \pm 2, \pm 3, ...)$ become weakly allowed. Since for these forbidden transitions, $|\Delta m_S| > 1$ or $|\Delta m_I| > 0$, they are often called multiple-quantum transitions. Note, however, that such a transition, where only *one* photon is absorbed or emitted, is still a *single-photon* process.

During a single-photon transition, the spin system absorbes or emits one photon, so that the angular momentum is conserved for a radiation field *perpendicular* to the quantization direction of the spin. During an electron spin transition with a positive g value or a nuclear spin transition with a negative g_n value, the absorbed or emitted photon is a σ^+ photon (Fig. 1a). In the rare case of a negative g value [19], or for a nucleus with positive g_n, the absorbed or emitted photons are σ^- photons (Fig. 1b).

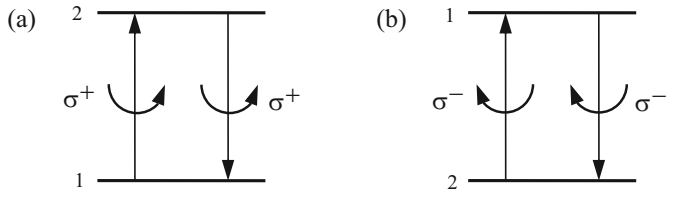


Fig. 1: Energy level diagram for a single-photon transition in a two-level system, fulfilling energy and angular momentum conservation (curved arrows symbolize the polarization of the absorbed or emitted photon). (a) $\epsilon(m_S = 1/2) > \epsilon(m_S = -1/2)$: One σ^+ photon is absorbed or emitted. (b) $\epsilon(m_S = -1/2) > \epsilon(m_S = 1/2)$: Order of the energy levels is reversed, one σ^- photon is absorbed or emitted.

In magnetic resonance *linear* radiation fields are used with a very few exceptions [19–21]. Such a linear field consists of both σ^- and σ^+ photons, since it can be considered as a superposition of a left and a right circular field. The generation of a circular field is experimentally more demanding, but has the advantage that the power needed to generate the same circular field amplitude ω_1 is reduced to one quarter compared to a linear field, and that the Bloch-Siegert shift (see below) is absent. Note, however, that for many of the multiple-photon processes described in this Lecture Notes, both σ^- and σ^+ photons are required, and transitions can not be induced by a circular radiation field.

2.1 Multiple-photon transitions in spin systems with unequally spaced energy levels

In EPR spectroscopy multiple-photon transitions, where the magnetic quantum number changes by $\Delta m_S > 1$, have been observed in various spin systems with unequally spaced energy levels (see for example [22–26]). In this kind of multiple-photon transitions, which requires real intermediate energy levels, the final state is reached without directly altering the populations of the intermediate states.

For the description of these multiple-photon transitions, higher-order time-dependent perturbation theory is usually applied. It has been found in cw EPR (and cw NMR), that (a) the unsaturated linewidth of an n-photon transition is $1/n$ of the linewidth of the single-photon transition, (b) in the absence of saturation effects the intensity of the signal for an n-photon transition is proportional to ω_1^{2n-1}, (c) the intensity of an n-photon transition between levels with energy ϵ_a and ϵ_b depends on the position of the intermediate energy levels with respect to the $(n-1)$ equally spaced virtual levels between ϵ_a and ϵ_b [27].

One radiation field perpendicular to $\mathbf{B_0}$

In this type of multiple-photon transitions, two or more σ^+ photons of the *same* frequency are absorbed when the energy levels are almost equally spaced. In EPR these transitions can be observed either with a linear or a right circular radiation field perpendicular to $\mathbf{B}_0$ (Fig. 2a).

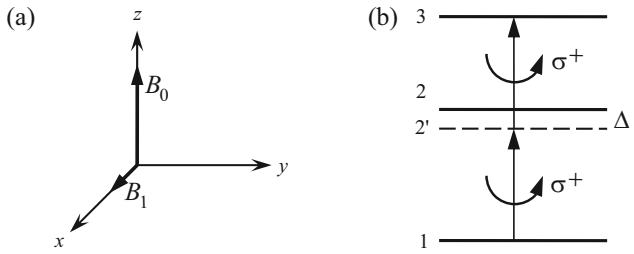


Fig. 2: (a) Field configuration in the laboratory frame and, (b) energy level diagram for a two-photon transition induced in a three-level system with unequally spaced energy levels by one right circular radiation field perpendicular to $\mathbf{B}_0$ (σ^+ photons).

As an example, we consider the simple situation of a right circular radiation field applied to a spin system consisting of three energy levels labeled by 1, 2, and 3 (Fig. 2b). In addition to the two single-photon transitions with $\Delta m_S = \pm 1$ between level 1 and 2, and level 2 and 3, a *two-photon transition* can be induced between level 1 and 3. If the states are not mixed the transition amplitude of the single-photon transition with $\Delta m_S = \pm 2$ is zero. For the two-photon transition, where two σ^+ photons of the same frequency $\omega_{mw} = \omega^{(13)}/2$ are absorbed, the transition

amplitude is given by [28]

$$a_{13} \propto \omega_1^2 \frac{< \phi_1 \mid S^+ \mid \phi_2 >< \phi_2 \mid S^+ \mid \phi_3 >}{\omega^{(32)} - \omega^{(31)}/2} \ , \tag{2}$$

with $\omega^{(32)} = \epsilon_3 - \epsilon_2$, $\omega^{(31)} = \epsilon_3 - \epsilon_1$, and the raising operator $S^+ = S_x + iS_y$. Such a two-photon transition can be considered as a forbidden transition between level 1 and 3 with $\Delta m_S = \pm 2$, carried out in two allowed steps, first from level 1 to an intermediate (virtual) level 2', and then from level 2' to level 3. The transition amplitude a_{13} in Eq. (2) is proportional to the square of the mw field amplitude ω_1 and inversely proportional to $\Delta = \omega^{(32)} - \omega^{(31)}/2$, where Δ is the energy difference between level 2 and the virtual level 2' . Thus, to observe a two-photon transition with sufficient intensity, the amplitude of the pumping field has to be large and Δ has to be small. Corresponding formulas hold for transitions where three or more photons are absorbed. For multiple-photon transitions in spin systems with unequally spaced energy levels, the number of absorbed σ^+ photons is equal to the total change in m_S of the electron spin, so that angular momentum is conserved.

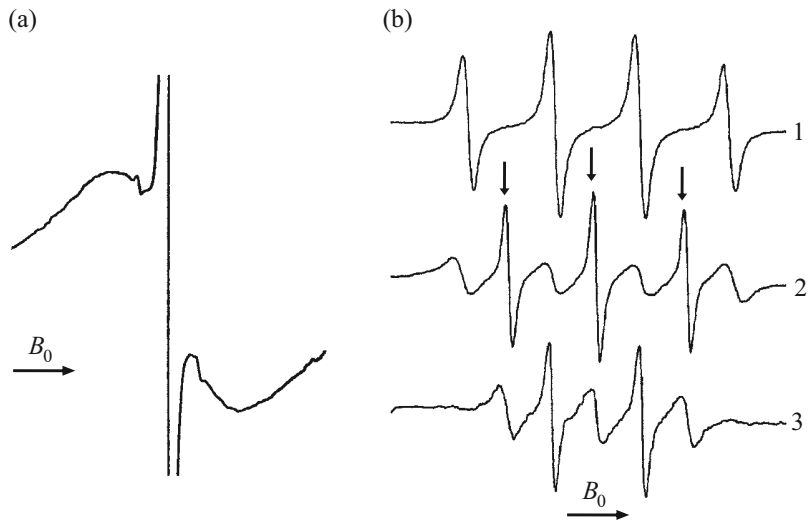


Fig. 3: Multiple-photon transitions in spin systems with unequally spaced energy levels. (a) Two-photon transition (narrow center line) of Ni^{+2} in a MgO single crystal (adapted from [29]). (b) Multiple-photon transitions of atomic oxygen. 1: mw power 0.6 mW; 2: 4.5 mW (arrows mark two-photon transitions); 3: 42 mW (adapted from [30]).

Two examples of multiple-photon transitions observed in electron spin systems with unequally spaced energy levels are given in Fig. 3. The spectrum of a Ni^{2+}-doped MgO single crystal with $S = 1$ is shown in Fig. 3a. When the spectrum is recorded with high mw power a broad single-photon transition superimposed by

a narrow double-photon line is observed [29]. The broadening of the single-photon line is caused by lattice strains, which shift the state $|0\rangle$ by different amounts relative to the states $|-1\rangle$ and $|1\rangle$. Since for the two-photon transition state $|0\rangle$ plays only the role of an intermediate level, the line remains narrow.

The second example (Fig. 3b) shows the spectral changes observed in atomic oxygen in the gas phase as a function of the mw power [30]. At low power (0.6 mW) only single-photon transitions are observed. But already at a power of 4.5 mW the spectrum is dominated by three *two-photon* transitions (marked by arrows). At a power of 42 mW the two *three-photon* transitions are the most intense lines, while the single-photon transitions are buried in the noise.

In all these EPR experiments where the multiple-photon transitions are *directly* observed, the line intensity is proportional to ω_1^{2n-1}. The situation is different when the multiple-photon transitions are *indirectly* detected via the change in polarization of another transition. This is the case, for example, in cw electron-nuclear double resonance (ENDOR) experiments where nuclear frequencies are detected via desaturation of an EPR line. For a non-saturating rf field the intensity of an n-photon transition is then proportional to ω_1^{2n}.

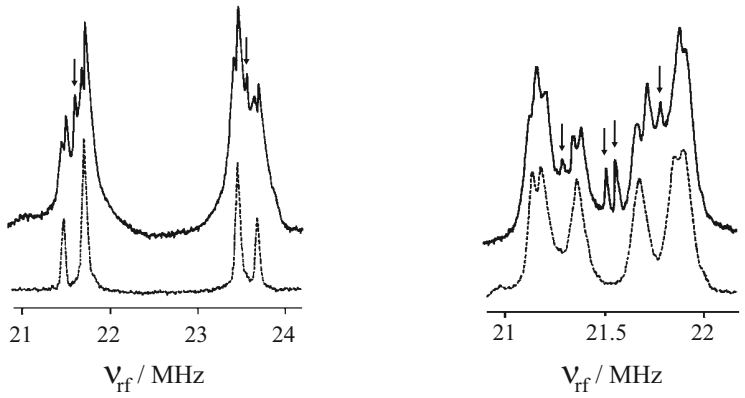


Fig. 4: Two-photon cw ENDOR transitions observed in bis(salicylaldoximato)copper(II), $Cu(sal)_2$, diluted into a $Ni(sal)_2$ single crystal. Two different settings of the crystal are shown. Bottom: low rf power, no two-photon transitions are observed; top: high rf power, arrows denote two-photon transitions (adapted from [28]).

Fig. 4 shows two examples for the observation of two-photon ENDOR transitions (marked by arrows) in a copper complex with two magnetically equivalent nitrogen ligands [28]. The intensity of the two-photon transitions depends on the parameter Δ which can be varied via the orientation-dependent nuclear quadrupole interaction by rotating the crystal.

Two radiation fields perpendicular to $\mathbf{B}_0$

The restriction $\omega^{(32)} \approx (\epsilon_3 - \epsilon_1)/2$ for the intermediate energy level may be dropped by using a *bichromatic* field perpendicular to $\mathbf{B}_0$ with frequencies $\omega_{\mathrm{mw},1}$ and $\omega_{\mathrm{mw},2}$ and amplitudes $\omega_1 = -\gamma_e B_1$ and $\omega_2 = -\gamma_e B_2$, as is shown in Fig. 5a.

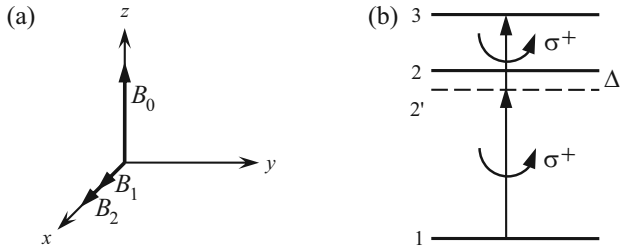


Fig. 5: (a) Field configuration in the laboratory frame and, (b) energy level diagram for a two-photon transition induced in a three-level system with unequally spaced energy levels by two right circular radiation fields perpendicular to $\mathbf{B}_0$ (σ^+ photons).

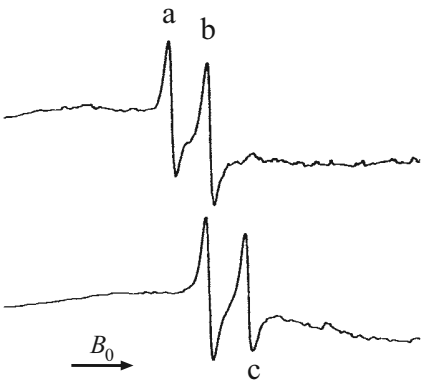


Fig. 6: Two-photon transitions of Ni^{+2} in a MgO single crystal induced by two mw fields of different frequencies (adapted from [31]).

In this scheme, $\omega_{\mathrm{mw},1}$ and $\omega_{\mathrm{mw},2}$ fulfilling the resonance condition $\omega_{\mathrm{mw},1} + \omega_{\mathrm{mw},2} = \omega^{(31)}$, may be chosen such that $\Delta = \omega^{(21)} - \omega_{\mathrm{mw},1}$ is small for any value of ϵ_2. The transition amplitude for this type of multiple-photon transitions is then given by

$$a_{13} \propto \omega_1 \omega_2 \frac{< \phi_1 \mid S^+ \mid \phi_2 >< \phi_2 \mid S^+ \mid \phi_3 >}{\omega^{(21)} - \omega_{\mathrm{mw},1}} \ . \tag{3}$$

Again corresponding formulas hold for transitions where three or more photons are absorbed.

An example for a two-photon transition induced by two mw frequencies is shown in Fig. 6 [31]. The system is again a Ni^{2+}-doped MgO single crystal. Line (a) or (c) is observed when the absorption of mw with frequency $\omega_{\mathrm{mw},1}$ or $\omega_{\mathrm{mw},2}$ is recorded. Line (b) in the center of the spectrum is attributed to the two-photon transition, where one photon of each radiation field is absorbed.

2.2 Multiple-photon transitions in two-level systems

Multiple-photon transitions can also be induced between two energy levels when no real intermediate level exist. In a two-level system, or a multi-level system where only two levels are involved in the experiment, one has to distinguish between multiple-photon processes where either an *odd* or an *even* number of photons is absorbed.

One linear radiation field

Consider a two-level system with electron spin $S = 1/2$ interacting with a linear radiation field with frequency ω_{mw}. For an angle θ between this field and the static field $\mathbf{B}_0$, the laboratory frame Hamiltonian is given by

$$\mathcal{H}(t) = \omega_S S_z + 2\omega_1 \cos(\omega_{\mathrm{mw}} t)\,[\cos(\theta) S_z + \sin(\theta) S_x] \ , \tag{4}$$

with the electron Zeeman frequency $\omega_S = -\gamma_e B_0$. The two eigenstates of the unperturbed spin system are denoted by α and β, with energies $\epsilon_\alpha = \omega_S/2$ and $\epsilon_\beta = -\omega_S/2$.

To follow the evolution of the spin system under the Hamiltonian in Eq.(4) the Schrödinger equation for the corresponding evolution operator has to be solved. To eliminate the time dependence of the Hamiltonian, Shirley implemented the Floquet theorem [15] as is outlined in the Appendix. Following this approach the transition probability between the two eigenstates of the time-dependent Hamiltonian $\mathcal{H}(t)$ can be found by using the corresponding Floquet Hamiltonian with matrix elements

$$\langle \alpha n | \mathcal{H}_{\mathrm{F}} | \beta m \rangle = \mathcal{H}^{n-m}_{\alpha\beta} + n\omega_{\mathrm{mw}} \delta_{\alpha\beta} \delta_{nm} \ , \tag{5}$$

where $\langle \alpha n|$ represent the Floquet states, α describes the spin state and n denotes the component in the Fourier expansion $\mathcal{H}(t) = \sum_n \mathcal{H}^n_{\alpha\beta} e^{in\omega_{\mathrm{mw}} t}$. The Hamiltonian in Eq. (5) has non-zero off-diagonal elements, which couple states that differ in the number of photons by $\Delta n = 1$ and in the spin projections by $\Delta m_S = 0, \pm 1$. The possible transitions can be revealed by checking the behavior of the energy levels in the vicinity of the $(n-m)$-resonance conditions $\epsilon_\alpha - \epsilon_\beta = (n-m)\omega_{\mathrm{mw}}$. The corresponding energy levels $\epsilon_{\alpha n} = \epsilon_\alpha + n\omega_{\mathrm{mw}}$ and $\epsilon_{\beta m} = \epsilon_\beta + m\omega_{\mathrm{mw}}$ are degenerated when the oscillating field is zero. For $\omega_1 \neq 0$, the level crossings $\epsilon_{\alpha n} = \epsilon_{\beta m}$ are lifted. The existence of multiple-photon transitions thus implies a level anticrossing, and *vice versa*.

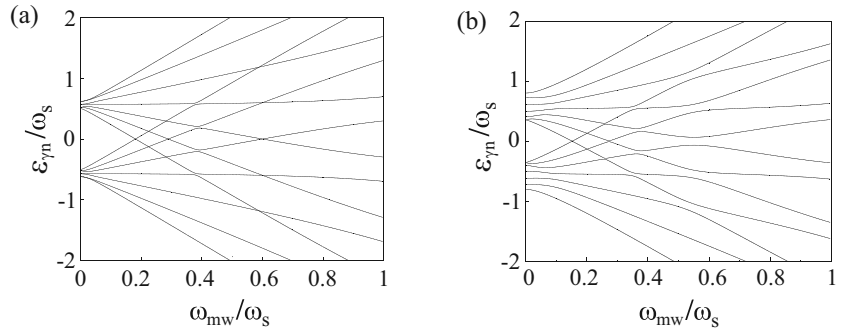


Fig. 7: Eigenvalues of the Floquet Hamiltonian calculated as a function of the normalized photon energy ω_{mw}/ω_S for $\omega_1/\omega_S = 0.2$ and two angles θ between $\mathbf{B}_0$ and $\mathbf{B}_1$. (a) Eigenvalues for $\theta = 90°$. (b) Eigenvalues for $\theta = 45°$.

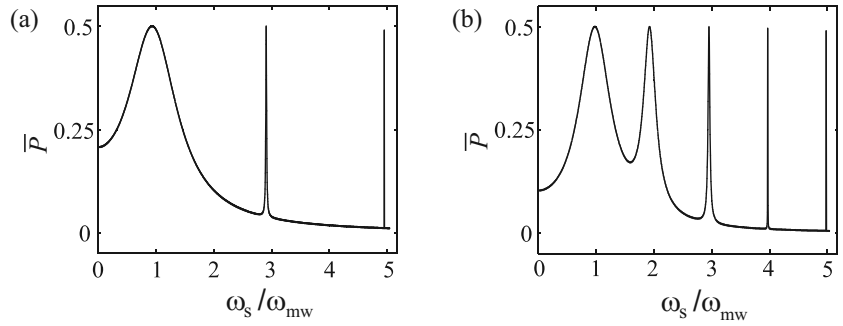


Fig. 8: Time-averaged transition probabilities as a function of the normalized Larmor frequency ω_S/ω_{mw} for $\omega_1/\omega_S = 0.25$. (a) Radiation field perpendicular to $\mathbf{B}_0$. (b) Direction of the radiation field tilted by $\theta = 45°$ (adapted from [32]).

To demonstrate the existence of anticrossings, the eigenvalues of $\mathcal{H}_F$ have been calculated as a function of ω_{mw}/ω_S with $\theta = 90°$ and 45°, and $\omega_1/\omega_{mw} = 0.25$ (Figs. 7a and b). Anticrossings are found in the vicinity of $\omega_{mw}/\omega_S = 1/n$. For an oscillating field perpendicular to $\mathbf{B}_0$ an anticrossing is observed at $\omega_{mw}/\omega_S = 0.38$, corresponding to the position of a three-photon resonance (Fig. 7a). For $\theta = 45°$, anticrossings occur at ω_{mw}/ω_S =0.55, 0.37 and 0.26; i.e., at two-, three- and four-photon resonances (Fig. 7b). A radiation field at an angle of 45° to $\mathbf{B}_0$ may thus induce both odd *and* even resonances.

Numerical computations of the time-averaged transition probability performed for $\mathcal{H}_F$ as a function of the normalized Larmor frequency ω_S/ω_{mw} are plotted in Figs. 8a and b for $\theta = 90°$ and $45°$. For a radiation field with two frequencies, the analysis is similar.

One linear radiation field perpendicular to $\mathbf{B}_0$

In a two-level system with electron spin $S = 1/2$ and energy levels 1 and 2, multiple-photon transitions with an *odd* number of photons can be induced by a linear radiation field *perpendicular* to $\mathbf{B}_0$ (Fig. 9a). In a $(2n+1)$-photon transition, $(n+1)$ σ^+ photons and $n\,\sigma^-$ photons are absorbed, so that the angular momentum is conserved.

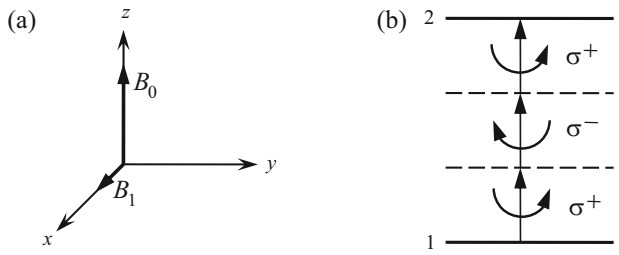


Fig. 9: (a) Field configuration in the laboratory frame and, (b) energy level diagram for a three-photon transition induced between the two states of an $S = 1/2$ spin system by a linear radiation field perpendicular to $\mathbf{B}_0$ (σ^+ and σ^- photons).

This type of multiple-photon transitions can only be induced by a *linear* field which consists of both σ^+ and σ^- photons. The energy level scheme for a three-photon transition $\sigma^+ + \sigma^- + \sigma^+$ with $\omega_{mw} = \omega_S/3$, where two σ^+ photons and one σ^- photon are absorbed, is given in Fig. 9b. For this process the transition amplitude is found to be

$$a_{12}^{(3)} = \omega_{eff}^{(3)} = \omega_1^3/2\omega_{mw}{}^2 \,. \tag{6}$$

The observation of such multiple-photon resonances has been reported for Cr^+-doped gallium phosphide [33]. Figure 10 shows the X-band cw EPR spectrum induced by red light.

In addition to the single-photon transition at 328 mT, also the two-photon transition at 656 mT and the three-photon transition at 984 mT (arrow) are induced. At high mw power multiple-photon transitions with n up to 7 could be observed. It is assumed that the very long relaxation time of this material allows the observation of the multiple-photon transitions. In this experiment with the mw field oriented perpendicular to $\mathbf{B}_0$, not only odd resonances as predicted by theory, but also even resonances could be observed (see below).

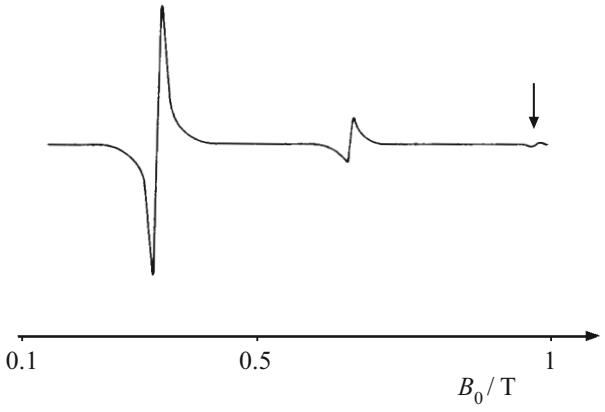


Fig. 10: Light-induced EPR spectrum of chromium-doped gallium phosphide showing a single-photon, a two-photon, and a three-photon transition (adapted from [33]).

One linear radiation field at arbitrary orientation

A linear radiation field, which includes an angle $\theta < 90°$ with the static field $\mathbf{B}_0$, can be decomposed into a linear field component *perpendicular* to $\mathbf{B}_0$ consisting of σ^- and σ^+ photons, and a field component *parallel* to $\mathbf{B}_0$ consisting of π photons (Fig. 11a) [34]. Consequently, multiple-photon transitions with both an *even* and an *odd* number of absorbed photons can be observed, as is shown in Fig. 11b for the transitions $\sigma^+ + \pi$ and $\sigma^+ + 2 \times \pi$.

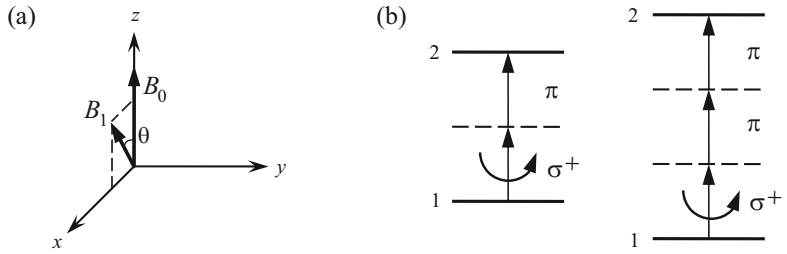


Fig. 11: (a) Field configuration in the laboratory frame and, (b) energy level diagram for a two-photon and a three-photon transition induced between the two states of an $S = 1/2$ spin system by a linear radiation field which includes an angle $\theta < 90°$ with $\mathbf{B}_0$ (σ^+, σ^-, and π photons).

For a two-photon transition induced by a radiation field with frequency $\omega_{\mathrm{mw}} = \omega_S/2$, the transition amplitude given by

$$a_{12} \propto \frac{\omega_1^2 \, \sin(2\theta)}{2\omega_S} \tag{7}$$

is maximum for $\theta = 45°$.

First observations of multiple-photon transitions of the type $\sigma^+ + k \times \pi$ have been reported by Winter in an optical pumping experiment [7]. The observation of *even* resonances in the spectrum of Cr^+-doped gallium phosphide mentioned above (Fig. 10) is explained by a distortion of the mw field caused by the high dielectric constant of the sample, which results in a component of the radiation field along $\mathbf{B}_0$.

The so-called *second-harmonic-detected EPR* experiments described by Boscaino and coworkers [35–39] are also based on two-photon transitions of the type $\sigma^+ + \pi$. In this approach a bimodal cavity with two resonant modes in the frequency range 2-6 GHz is used. In the first mode, a linear radiation field with frequency $\omega_{\mathrm{mw}} \approx \omega_S/2$, which includes with $\mathbf{B}_0$ an angle of $\theta = 45°$, induces two-photon transitions in an $S = 1/2$ spin system. The resulting second-harmonic signal at the Larmor frequency ω_S is detected via the second mode perpendicular to $\mathbf{B}_0$. This scheme has the advantage that the signal at frequency ω_S can be recorded *during* excitation, since the radiation field at half this frequency is far off-resonant from the observation frequency. The experimental scheme has been applied in various investigations of two-level systems (see for example [40]). Very recently, second-harmonic detection has been used to study the decay properties of stimulated nutation echoes [41].

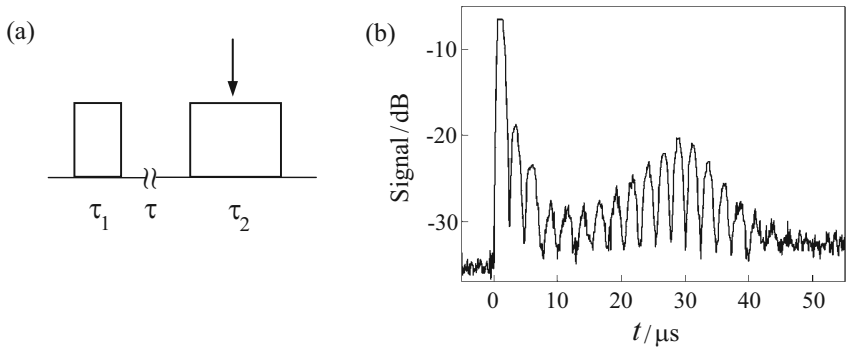


Fig. 12: Stimulated nutation echo induced by a linear radiation field ($\theta = 45°$ and recorded via the second-harmonic signal. (a) Pulse sequence, arrow marks echo maximum. (b) Stimulated nutation echo in γ-irradiated quartz glass (adapted from [41]).

The two-pulse sequence used to generate this type of echoes is shown in Fig. 12a. Both pulses with frequency $\omega_{\mathrm{mw}} \approx \omega_S/2$ drive the two-photon transition

$\sigma^+ + \pi$. The first pulse of length τ_1 burns a polarization pattern into an inhomogeneous line [18], which is recalled by the second pulse after a waiting time $\tau \gg \tau_1$. The stimulated nutation echo is formed at time $t = \tau_1$ after the onset of the second pulse (Fig. 12b [41]).

Two or more radiation fields perpendicular to $\mathbf{B}_0$

Multiple-photon processes become more complex when transitions are induced in a two-level system by linear bichromatic or tetrachromatic radiation fields perpendicular to $\mathbf{B}_0$. Multiple-photon transitions with an *odd* number n of photons can be induced by *two* linear radiation fields with frequencies $\omega_{\mathrm{mw},1}$ and $\omega_{\mathrm{mw},2}$ and amplitudes $\omega_1 = \omega_2$ (Fig. 13a), when the resonance condition

$$\omega_S = \omega_{\mathrm{mw},1} \pm [n^2(\omega_{\mathrm{mw},2} - \omega_{\mathrm{mw},1})^2 - \omega_1^2]^{1/2} \tag{8}$$

is fulfilled [42]. A three-photon transition of this type is shown in Fig. 13b.

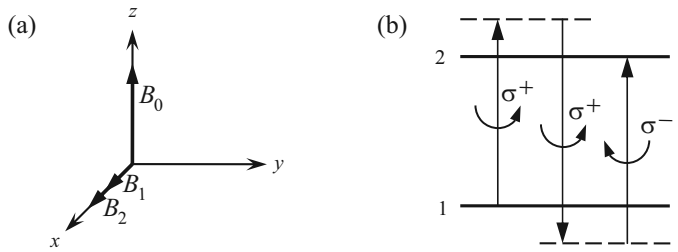


Fig. 13: (a) Field configuration in the laboratory frame and, (b) energy level diagram for a three-photon transition induced between the two states of an $S = 1/2$ spin system by two linear radiation fields perpendicular to $\mathbf{B}_0$ (σ^+ and σ^- photons).

Recently, Hyde and co-workers introduced a multiple-photon EPR experiment, where the spin system is excited by two (or four) cw radiation fields with closely spaced mw frequencies [43–47]. In the case of two radiation fields with frequencies $\omega_{\mathrm{mw},1} = \omega_{\mathrm{mw}} + \frac{1}{2}\Delta\omega$ and $\omega_{\mathrm{mw},2} = \omega_{\mathrm{mw}} - \frac{1}{2}\Delta\omega$ and amplitudes ω_1, ω_2, the multiple-photon process creates coherences that manifest as sidebands at frequencies $\omega_{\mathrm{mw}} \pm (k + \frac{1}{2})\Delta\omega$. Corresponding experiments have also been reported in pulse NMR [48, 49] and cw NMR [12, 50, 51], and in the optic regime [52].

Each individual sideband can be down-converted with respect to the reference frequency ω_{mw} and detected with a phase-sensitive detector. The same detection scheme has also been used in cw electron–electron double resonance [53] and cw ENDOR [54] experiments.

This approach can be used to measure pure absorption EPR spectra [55]. As an example, in Fig. 14 the conventional cw EPR spectrum of nitrous oxide reductase is compared with the three-photon EPR spectrum recorded without field modulation, and the corresponding first-harmonic spectrum obtained by pseudo-modulation [56]. The difference between the two spectra (marked by an arrow) is

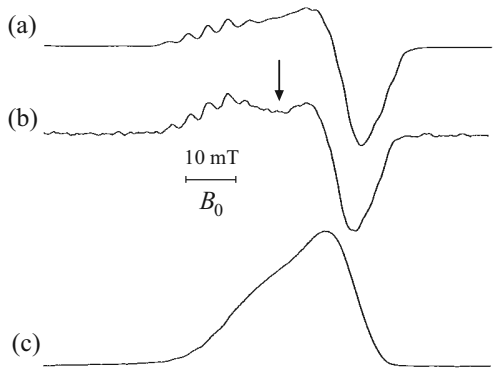


Fig. 14: EPR on nitrous oxide reductase substituted with ^{63}Cu, temperature 20 K. (a) Cw EPR spectrum. (b) First harmonic of the spectrum in (c), using pseudomodulation. (c) Three-photon EPR spectrum (adapted from [55]).

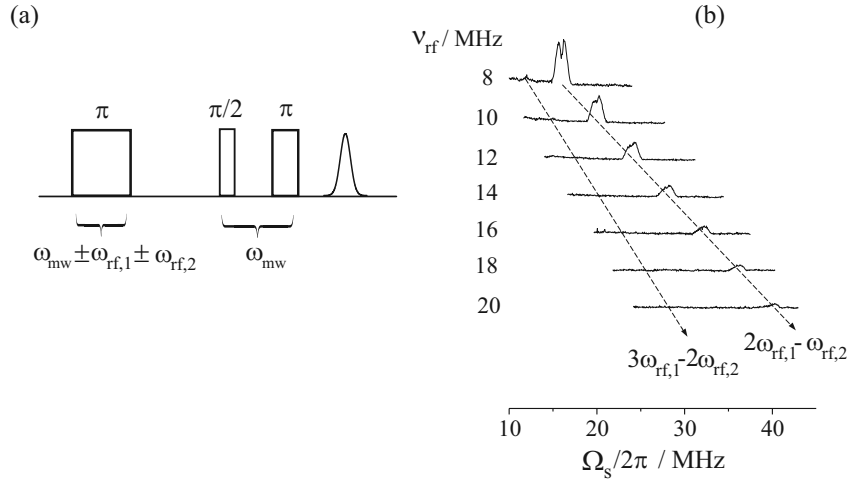


Fig. 15: Three- and five-photon resonances in γ-irradiated quartz glass at X-band frequency. (a) Inversion recovery pulse sequence with a tetrachromatic inversion pulse. (b) Multiple-photon resonances as a function of $\omega_{rf,2}$ at different fixed frequencies $\omega_{rf,1}$ of the tetrachromatic pulse. Three-photon resonances are observed for $2\omega_{rf,1} - \omega_{rf,2} = 0$ (dashed line). Very weak five-photon resonances with $3\omega_{rf,1} - 2\omega_{rf,2} = 0$ (dashed line) can be observed at the two top traces. For convenience, the spectra are inverted.

traced back to the enhanced suppression of the forbidden transitions in the multiple-photon spectrum.

This type of multiple-photon process can also be studied with pulse excitation. Recently, an inversion recovery experiment with the pulse sequence shown in Fig. 15a and a tetrachromatic radiation field has been carried out on γ-irradiated quartz glass.

The tetrachromatic pulses were created by mixing the mw frequency ω_{mw} with two radio frequencies $\omega_{\mathrm{rf},1}$ and $\omega_{\mathrm{rf},2}$. An odd number of photons is absorbed from the tetrachromatic radiation field, which induces up-down transitions and creates polarization at frequency ω_S, corresponding to the center frequency ω_{mw} of the inversion pulse. The resonance condition, $\omega_S = \omega_{\mathrm{mw}} + n_1\omega_{\mathrm{rf},1} + n_2\omega_{\mathrm{rf},2}$, is fulfilled for example for the three-photon transition with n_1=2 and n_2= -1, and the five-photon transition with n_1=3 and n_2= -2. The integrated echo intensities as a function of $\omega_{\mathrm{rf},2}$ at different fixed frequencies $\omega_{\mathrm{rf},1}$ are plotted in Fig. 15b. In addition to the three-photon resonances, weak five-photon resonances become visible with radio frequencies of 8 and 10 MHz.

3 Effects of oscillating longitudinal field

In this section we describe multiple-photon transitions induced by a *bichromatic* radiation field consisting of a linear or right circular mw field in the gigahertz range, *perpendicular* to $\mathbf{B}_0$, and an rf field in the megahertz range, *parallel* to $\mathbf{B}_0$ (Fig. 16a). Such a bichromatic field induces multiple-photon transitions of the type $\sigma^+_{\mathrm{mw}} + k \times \pi_{\mathrm{rf}}$. In this section we understand by a bichromatic field always this kind of radiation field, and for the multiple-photon transitions we use the abbreviated form $\sigma^+ + k \times \pi$.

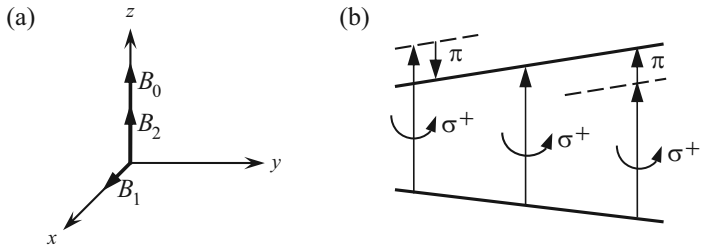


Fig. 16: (a) Field configuration in the laboratory frame and, (b) energy level diagram for the experiments with one mw field perpendicular to $\mathbf{B}_0$ and one rf field parallel to $\mathbf{B}_0$, showing two-photon resonances with $\omega_S = \omega_{\mathrm{mw}} \pm \omega_{\mathrm{rf}}$.

Two-photon transitions $\sigma^+ + \pi$ of this type (Fig. 16b) have first been reported by Winter in 1958 [34] on the free radical DPPH, and three years later by Burget and coworkers and Hashi on DPPH and other two-level systems [57, 58].

Recently, such two-photon transitions has also been observed in electron-nuclear multi-level systems, where they appear in cw ENDOR spectra at the corresponding hyperfine frequencies [59].

3.1 The field modulation used in cw EPR as a multiple photon process

In this section we demonstrate that field modulation commonly used in cw EPR spectroscopy, together with the mw field, represents a bichromatic radiation field which gives reason to complex multiple photon processes.

Since the dawn of cw EPR spectroscopy, modulation of the static magnetic field B_0 with subsequent phase-sensitive detection is used to improve signal-to-noise. As a consequence of this field modulation the *derivative* of the absorption signal is observed. When the modulation amplitude or the modulation frequency are larger than the width of a particular line, the line shape is distorted. The former situation is known as modulation broadening due to overmodulation, and in the latter case sidebands appear in the spectrum [60].

A first theoretical description of modulation effects was given by Karplus [61] for mw rotational spectroscopy, where either the excitation frequency or the energy levels of the molecule are modulated. In magnetic resonance this theory was used by Smaller [62], and was further developed by several authors to obtain quasi steady-state solutions of the modified Bloch equations [63–67]. These equations [67] give a correct description for any harmonics of the cw EPR signal, including distortions due to overmodulation.

In the classical approach the modulation of B_0 with frequency ω_{rf} , leads to the time-dependent field $B(t) = B_0 + 2B_2\cos(\omega_{\mathrm{rf}}t)$. The corresponding signal can be expanded to a Taylor series

$$S(t) \propto A(B_0) + \sum_{n=1}^{\infty} \frac{1}{n!} \frac{\mathrm{d}^n A(B_0)}{\mathrm{d}B_0^n} [2B_2\cos(\omega_{\mathrm{rf}}t)]^n \ , \tag{9}$$

where $A(B_0)$ is the absorption spectrum (Fig. 17a). For modulation amplitudes small compared to the linewidth, only the first-order term with $n = 1$ has to be considered. The first-harmonic absorption spectrum, the part of the absorption signal oscillating with frequency ω_{rf}, then consists of lines which are the first derivative $\mathrm{d}A(B_0)/\mathrm{d}B_0$ of the absorption lines (Figs. 17b).

With increasing modulation amplitude higher-order terms of the Taylor series in Eq.(9) become relevant (Fig. 17c) [68–70]. For modulation amplitudes large compared to the linewidth, the spacing between the positive and negative peak maximum of the distorted line corresponds to the modulation amplitude $2B_2$. This effect is routinely used to calibrate the modulation coils of EPR spectrometers.

We now discuss two semi-classical approaches, which in contrast to the modified Bloch equations can also be applied to pulse experiments.

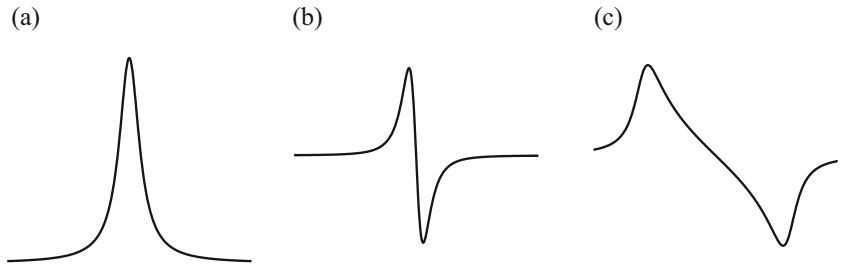


Fig. 17: (a) Lorentzian absorption line with width Γ. (b) First-harmonic line, modulation amplitude Γ/20 (first derivative of the absorption line). (c) First-harmonic line with strong overmodulation, modulation amplitude 6Γ.

Multiply tilted rotating frames

The influence of an rf field parallel to $\mathbf{B}_0$ on a spin system can be described by a series of transformations to tilted rotating frames [11, 32, 71]. This approach is restricted to cases where higher-order effects caused by strong radiation fields can be neglected.

An $S = 1/2$ electron spin system subject to an external field $\mathbf{B}_0$ and a bichromatic radiation field can be described by the Hamiltonian

$$\mathcal{H}_{\text{lab}}(t) = \omega_S S_z + 2\omega_1 \cos(\omega_{\text{mw}} t) S_x + 2\omega_2 \cos(\omega_{\text{rf}} t) S_z \ , \tag{10}$$

where $\omega_2 = -\gamma_e B_2$ is the amplitude of the radio frequency in angular frequency units. In a first step the Hamiltonian in Eq.(10) is transformed to a frame rotating with frequency ω_{mw}. Omitting the counter-rotating component of the mw field we find

$$\mathcal{H}_{\text{SRF}} = \Omega_S S_z + \omega_1 S_x + 2\omega_2 \cos(\omega_{\text{rf}} t) S_z \ , \tag{11}$$

with the resonance offset $\Omega_S = \omega_S - \omega_{\text{mw}}$. The Hamiltonian in this singly rotating frame is shown graphically in Fig. 18a. The effective field vector $\boldsymbol{\omega}_{\mathbf{eff}}$ and the z-axis include an angle $\alpha = \arctan(\omega_1/\Omega_S)$. The frame is then tilted around the y-axis by this angle, resulting in a new z'-axis parallel to $\boldsymbol{\omega}_{\mathbf{eff}}$. Finally, this frame is transformed to a frame rotating with frequency ω_{rf} (Fig. 18b). In this doubly rotating frame the equilibrium magnetization lies on a cone with apex angle 2α. For small angles α, corresponding to $\omega_1 \ll \Omega_S$, the magnetization is approximatively oriented along to the z'-axis. After omitting the counter-rotating term and the remaining linear component, the Hamiltonian is given by

$$\mathcal{H}_{\text{TRF},1} = \Omega_{S,1} S_z + \omega_{1,1} S_x \ , \tag{12}$$

with the new resonance offset $\Omega_{S,1} = \sqrt{\omega_1^2 + \Omega_S^2} - \omega_{\text{rf}} \approx \Omega_S - \omega_{\text{rf}}$. For the effective field amplitude $\omega_{1,1}$ in the first tilted rotating frame we find $\omega_{1,1} = -\omega_2 \sin(\alpha)$. Near resonance ($\Omega_{S,1} \ll \omega_{\text{rf}}$), $\omega_{1,1}$ can be simplified to

$$a_{12} = \omega_{1,1} \approx \frac{-\omega_1 \omega_2}{\omega_{\text{rf}}} \ , \tag{13}$$

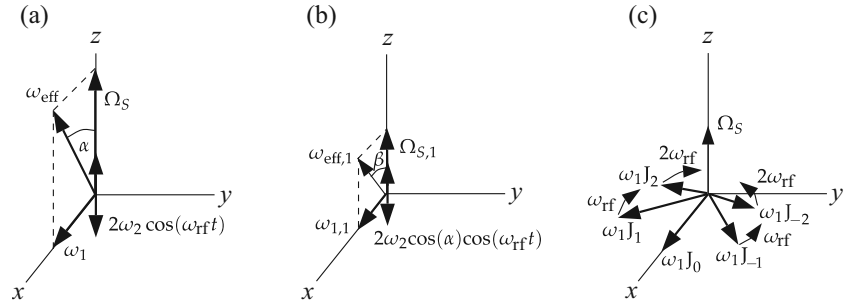


Fig. 18: Pictorial representation of the cw EPR Hamiltonian. (a) Singly rotating frame. (b) First tilted rotating frame, tilted from the z-axis by angle α and rotating with ω_{rf} (β is the tilt angle for the next transformation). (c) Toggling frame with $k = 0$ and $z = 2\omega_2/\omega_{\text{rf}}$.

leading to the Hamiltonian

$$\begin{aligned} \mathcal{H}_{\text{TRF},1} &= (\Omega_S - \omega_{\text{rf}}) S_z - \omega_{1,1}\, S_x \\ &= (\Omega_S - \omega_{\text{rf}}) S_z + \frac{\omega_1 \omega_2}{\omega_{\text{rf}}}\, S_x \ . \end{aligned} \tag{14}$$

Eq. (14) describes a two-photon transition, resonant with $\omega_S = \omega_{\text{mw}} + \omega_{\text{rf}}$.

By repeating this procedure, a Hamiltonian for any multiple-photon transition with frequency $\omega_S = \omega_{\text{mw}} + k\omega_{\text{rf}}$ can be derived. For the three-photon transition, with frequency $\omega_S = \omega_{\text{mw}} + 2\omega_{\text{rf}}$, for example, the Hamiltonian in the triply rotating frame is found to be

$$\mathcal{H}_{\text{TRF},2} = (\Omega_S - 2\omega_{\text{rf}}) S_z + \frac{\omega_1 \omega_2^2}{2{\omega_{\text{rf}}}^2}\, S_x \ . \tag{15}$$

The tilted frame approach delivers only correct results for rf amplitudes much weaker than the fields usually used in EPR experiments.

Toggling frames

The toggling frame approach does not suffer from this drawback [72, 73]. Starting with the Hamiltonian in the singly rotating frame (Eq. (11)), the time-dependent longitudinal component of the Hamiltonian is replaced by time-dependent transverse components using the rotation operator

$$R(t) = \mathrm{e}^{\mathrm{i}\left[k\omega_{\text{rf}} t + \left(\frac{2\omega_2}{\omega_{\text{rf}}}\right) \sin(\omega_{\text{rf}} t)\right] S_z} \ . \tag{16}$$

We then arrive at the toggling frame Hamiltonian

$$\begin{aligned}\mathcal{H}_{\mathrm{TF},k} &= R(t)\mathcal{H}_{\mathrm{SRF}}R^{-1}(t) - k\omega_{\mathrm{rf}}S_z - 2\omega_2\cos(\omega_{\mathrm{rf}}t)S_z \\ &= (\Omega_S - k\omega_{\mathrm{rf}})S_z + \omega_1 \mathrm{J}_{-k}(z)\, S_x \\ &\quad + \sum_{\substack{n=-\infty \\ n\neq -k}}^{\infty} \omega_1 \mathrm{J}_n(z)\; \mathrm{e}^{\mathrm{i}(k+n)\omega_{\mathrm{rf}}tS_z} S_x \mathrm{e}^{-\mathrm{i}(k+n)\omega_{\mathrm{rf}}tS_z}\,, \end{aligned} \tag{17}$$

with the normalized rf amplitude $z = 2\omega_2/\omega_{\mathrm{rf}}$. $\mathcal{H}_{\mathrm{TF},k}$ contains the resonance offset $(\Omega_S - k\omega_{\mathrm{rf}})$, a time-independent effective field amplitude $\omega_1 \mathrm{J}_{-k}(z)$, and the sum of the remaining time-dependent transverse perturbations. For a better understanding $\mathcal{H}_{\mathrm{TF}}$ is visualized in Fig. 18c for $k = 0$. The component $\omega_1\, \mathrm{J}_0(z)$ points along the x-axis, while all the other transverse components rotate around the z-axis with frequencies that are multiples of ω_{rf}. For a properly chosen value of k, the remaining resonance offset is minimum and the time-dependent terms of the Hamiltonian can be neglected in first-order. A multiple-photon transition of the type $\sigma^+ + k \times \pi$ has thus the transition amplitude

$$a_{12} = \omega_{1,k} = \omega_1\, \mathrm{J}_{-k}(z)\,. \tag{18}$$

Although the toggling frame is a convenient tool for the understanding of multiple-photon processes, it remains a *semiclassical* description.

Fully quantum mechanical description

For a fully quantized description of the cw EPR experiment with field modulation, only two modes of the radiation field, corresponding to the two frequencies, are of importance (see Appendix). All other frequencies are either strongly damped or far off-resonant. The fully quantized Hamiltonian is then given by

$$\begin{aligned}\mathcal{H} &= \omega_S S_z + \omega_{\mathrm{mw}}\left(a_{\mathrm{mw}}^{\mathrm{T}} a_{\mathrm{mw}} + \frac{1}{2}\mathbb{1}\right) + \omega_{\mathrm{rf}}\left(a_{\mathrm{rf}}^{\mathrm{T}} a_{\mathrm{rf}} + \frac{1}{2}\mathbb{1}\right) \\ &\quad + \frac{\omega_1}{\sqrt{N_{\mathrm{mw}}}}\left(a_{\mathrm{mw}}^{\mathrm{T}} + a_{\mathrm{mw}}\right) S_x + \frac{\omega_2}{\sqrt{N_{\mathrm{rf}}}}\left(a_{\mathrm{rf}}^{\mathrm{T}} + a_{\mathrm{rf}}\right) S_z\,. \end{aligned} \tag{19}$$

The function space is spanned by $|m_S, n, m\rangle = |m_S\rangle \otimes |n\rangle \otimes |m\rangle$, where n and m are the numbers of mw and rf photons.

Due to the low frequency of the mw and rf fields, $\mathcal{H}$ can be replaced by a Hamiltonian in a two-mode Floquet space, as is shown in Appendix A.1-3. This approach provides a simpler mathematical treatment of the problem and is used for further evaluation. However, it is important to mention that Fourier indices of the Floquet states correspond to the photon state occupation numbers of the fully quantized Hamiltonian in Eq.(19). This correspondence proves that the modulation sidebands discussed later are caused by *multiple-photon transitions*.

The resonance condition for a multiple-photon transition of the type $\sigma^+ + k \times \pi$ between the two levels $\langle \alpha, 0, 0|$ and $\langle \beta, 1, k|$ is fulfilled when $\langle \alpha, 0, 0|\mathcal{H}_{\mathrm{F}}|\alpha, 0, 0\rangle \approx \langle \beta, 1, k|\mathcal{H}_{\mathrm{F}}|\beta, 1, k\rangle$, or $\omega_S \approx \omega_{\mathrm{mw}} + k\,\omega_{\mathrm{rf}}$. Coupling between the degenerate levels will then lead to a coherent transition in the form of Rabi oscillations.

Fig. 19a shows a segment of the matrix representation of the two-mode Floquet Hamiltonian. The two levels $|\alpha, 0, 0\rangle$ and $|\beta, 1, 1\rangle$ are degenerate, corresponding to the two-photon transition $\sigma^+ + \pi$. Since there is no direct coupling between the elements, only second- and higher-order effects are relevant. The two pathways for the two-photon transition are indicated by arrows and correspond to the transitions I and II in Fig. 19b. The next possible transition pathways are those of the four-photon transitions, $-\sigma^+ + \pi + 2\sigma^+$, $2\sigma^+ + \pi - \sigma^+$, $-\pi + \sigma^+ + 2\pi$, and $2\pi + \sigma^+ - \pi$ (transitions III to VI in Fig. 19b).

Using a perturbation approach [74], the effective Hamiltonian for any two coupled Floquet state can be derived. For two-photon transitions, I and II in Fig. 19, the Hamiltonian is given by

$$\mathcal{H}_{\text{eff}} = \left[\Omega_S - \omega_{\text{rf}} + \frac{\omega_1^2}{2\omega_{\text{rf}}} + \frac{\omega_1^2}{4\omega_{\text{mw}}}\right] S_z + \omega_{1,1} S_x \ , \tag{20}$$

where $\Delta_{\text{BS}} = \omega_1^2/(4\omega_{\text{mw}})$ is the Bloch-Siegert shift [75]. The effective field amplitude $\omega_{1,1}$ is approximately $(-\omega_1\omega_2/\omega_{\text{rf}})$, a result which is also obtained with the tilted frame approach (Eq. (13)).

The perturbation approach or a numerical evaluation of the two-mode Floquet Hamiltonian provide solutions for all possible transitions and transition amplitudes. However, it is better to reduce the number of modes in the Floquet space to get a more compact but still complete presentation of related states and transitions.

We have already seen that the toggling frame is a convenient choice for a semiclassical description of $\sigma^+ + k \times \pi$ transitions. The parameter k of the toggling frame may be chosen arbitrarily and does not have to fulfill any kind of resonance condition, since the resonant transition is selected later. We simply set $k = 0$ and get, according to Eq. (17), the toggling frame Hamiltonian

$$\mathcal{H}_{\text{TF},0} = (\Omega_S + \Delta_{\text{BS}})\, S_z + \sum_{n=-\infty}^{\infty} \omega_1 \text{J}_n(z)\, \text{e}^{\text{i}n\omega_{\text{rf}}tS_z} S_x \text{e}^{-\text{i}n\omega_{\text{rf}}tS_z} \ . \tag{21}$$

From Eq. (21) the time-independent single-mode Floquet Hamiltonian $\mathcal{H}_{\text{F,TF}}$ is constructed. A small segment of the matrix representation is shown in Fig. 19c. The Hamiltonian $\mathcal{H}_{\text{F,TF}}$ has a direct coupling element between two degenerate levels, taking into account all pathways where *one* mw σ^+ photon is absorbed. The coherence is described in the function space of $\mathcal{H}_{\text{F,TF}}$. The back-transformation to the singly rotating frame splits this single coherence into a set of coherences corresponding to all involved virtual levels. For the effective Hamiltonian of a resonant $\sigma^+ + k \times \pi$ multiple-photon transition calculated from $\mathcal{H}_{\text{F,TF}}$ (see Fig. 19c), we find [72, 73]

$$\mathcal{H}_{\text{eff}} = ([\omega_S + \Delta_{\text{BS}} + \Delta_k] - [\omega_{\text{mw}} + k\omega_{\text{rf}}])\, S_z + \omega_{1,k} S_x \ , \tag{22}$$

with a Bloch-Siegert-like resonance shift (in second order)

$$\Delta_k = \left(\frac{2\omega_1^2}{\omega_{\text{rf}}}\right) \sum_{l \neq k} \frac{\text{J}_l^2(z)}{(k-l)} \ , \tag{23}$$

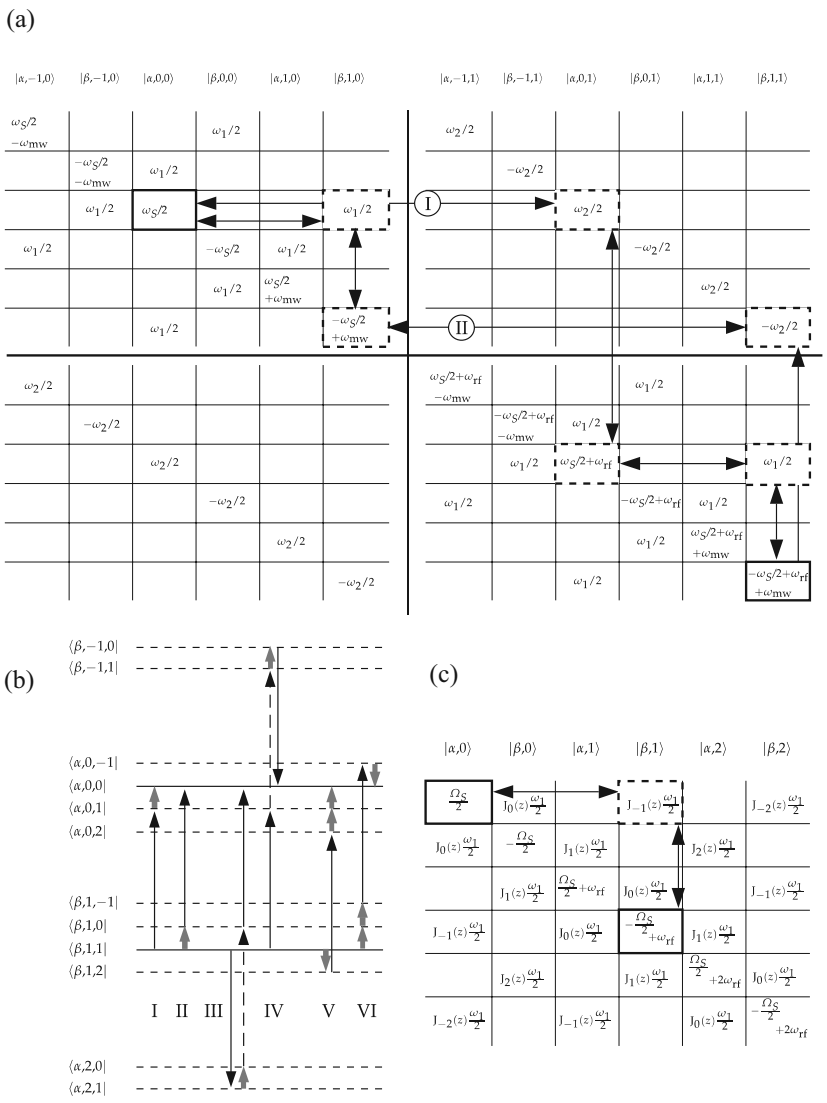


Fig. 19: (a) Segment of the two-mode Floquet space Hamiltonian $\mathcal{H}_F$. Full rectangles: Degenerate levels $|\alpha,0,0\rangle$ and $|\beta,1,1\rangle$. Dashed rectangles: Coupling elements and intermediate levels of the two-photon transition. The two pathways I and II are indicated by arrows. (b) Energy level scheme with transitions of the two two-photon pathways I and II and the four four-photon pathways III to VI. Black arrows: σ^+_{mw} photons, dashed arrows: σ^-_{mw} photons, grey arrows: π_{rf} photons. (c) Segment of the single-mode Floquet Hamiltonian $\mathcal{H}_{F,TF,0}$ (adapted from [72]).

and the effective field amplitude [2]

$$\begin{aligned}\omega_{1,k} &= \omega_1 \left[c_k^{(0)} + c_k^{(3)}\right] \\ &= \omega_1 \left[\mathrm{J}_{-k}(z) + \left(\frac{\omega_1}{2\omega_{\mathrm{rf}}}\right)^2 \sum_{l\neq k}\sum_{m\neq 0} \frac{\mathrm{J}_{-l}(z)\,\mathrm{J}_{m-l}(z)\,\mathrm{J}_{m-k}(z)}{(l-k)m}\right] . \end{aligned} \tag{24}$$

For $\omega_1 \ll \omega_{\mathrm{rf}}, \Delta_k$, the higher-order terms in Eq. (24) and the Bloch-Siegert shift Δ_{BS} can be neglected, resulting in

$$\mathcal{H}_{\mathrm{eff}} = \left(\Omega_S - k\omega_{\mathrm{rf}}\right) S_z + \omega_1\, \mathrm{J}_{-k}(z)\; S_x \; . \tag{25}$$

This expression is equivalent to the Hamiltonian derived from the semiclassical toggling frame approach in Eq. (17) after omitting the time-dependent perturbations.

Description of sidebands

The cw EPR absorption signal is proportional to the transverse component of the steady-state magnetization out-of-phase with the mw field. The steady-state solution is most easily obtained from the effective Hamiltonian given in Eq. (25) with the assumption that each $\sigma^+ + k \times \pi$ transition can be treated separately. The density operator at thermal equilibrium, $\sigma_{\mathrm{SRF}} = -S_z$, is transformed to the kth toggling frame by $\sigma_{\mathrm{TF},k}(t) = R(t)\sigma_{\mathrm{SRF}}(t)R^{-1}(t)$.

Since the detection operator $D = S_y$ is expressed in the singly rotating frame, the density operator has to be transformed back to this frame,

$$\begin{aligned}\sigma_{\mathrm{SRF}}(t) &= R^{-1}(t)\sigma_{\mathrm{TF},k}(t)R(t) \\ &= \sum_{n=-\infty}^{\infty} \mathrm{J}_n(z)\, \mathrm{e}^{-\mathrm{i}(k+n)\omega_{\mathrm{rf}}tS_z}\sigma_{\mathrm{TF},k}(t)\mathrm{e}^{\mathrm{i}(k+n)\omega_{\mathrm{rf}}tS_z} . \end{aligned} \tag{26}$$

The time-independent density operator in the toggling frame is then transformed to a series of terms in the singly rotating frame, oscillating with multiples of the modulation frequency. Summation over all possible multiple-photon resonances results in the general solution for the cw EPR absorption signal

$$\begin{aligned}\langle S_y\rangle &= tr[\sigma_{\mathrm{SRF}}\, S_y] = \sum_{n=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} \omega_1\, \mathrm{J}_n(z)\, \mathrm{J}_{-k}(z) \\ &\quad \times \frac{T_2\, \cos([k+n]\omega_{\mathrm{rf}}t) - (\Omega_S - k\omega_{\mathrm{rf}})\; T_2^2\, \sin([k+n]\omega_{\mathrm{rf}}t)}{1 + \omega_1^2\, \mathrm{J}_k^2(z)\; T_1\, T_2 + (\Omega_S - k\omega_{\mathrm{rf}})^2\; T_2^2} \; . \end{aligned} \tag{27}$$

Apart from saturation, Eq. (27) is identical to the formula derived from the Bloch equations [67]. Using Eq. (27), the absorption spectrum for any harmonic of the

[2] Note that the expression for the term $c_k^{(3)}$ in [72] contains a sign error.

modulation frequency can be calculated. For the zeroth-harmonic spectrum, only the constant terms are relevant

$$\langle S_y\rangle_0 \quad = \quad \sum_{k=-\infty}^{\infty} \frac{T_2\,\omega_1 \mathrm{J}_k^2(z)}{1+\omega_1^2\,\mathrm{J}_k^2(z)\;T_1\,T_2+(\Omega_S-k\omega_{\mathrm{rf}})^2\;T_2^2}\;. \tag{28}$$

In cw EPR the first-harmonic absorption spectrum, given by

$$\begin{aligned}\langle S_y\rangle_1 \quad = \quad \sum_{k=-\infty}^{\infty} \Bigg[& \frac{T_2\,\omega_1 \mathrm{J}_{-k}(z)\left[\mathrm{J}_{1-k}(z)+\mathrm{J}_{-1-k}(z)\right]}{1+\omega_1^2\,\mathrm{J}_k^2(z)\;T_1\,T_2+(\Omega_S-k\omega_{\mathrm{rf}})^2\;T_2^2}\;\cos(\omega_{\mathrm{rf}}t) \\ - & \frac{(\Omega_S-k\omega_{\mathrm{rf}})\;T_2^2\,\omega_1 \mathrm{J}_{-k}(z)\left[\mathrm{J}_{1-k}(z)-\mathrm{J}_{-1-k}(z)\right]}{1+\omega_1^2\,\mathrm{J}_k^2(z)\;T_1\,T_2+(\Omega_S-k\omega_{\mathrm{rf}})^2\;T_2^2}\;\sin(\omega_{\mathrm{rf}}t)\Bigg]\;,\end{aligned} \tag{29}$$

is usually measured. These equations becomes more clear from the plots shown in Fig. 20, which represent the field-swept EPR absorption spectra of the zeroth-harmonic and the first- and second-harmonic signal with the components in-phase and out-of-phase with respect to the rf field.

The zeroth-harmonic spectrum is symmetric (Fig. 20b) and consists of a center line (single-photon transition) and sidebands (multiple-photon transitions) with spacings $\Delta B = \omega_{\mathrm{rf}}/\gamma_e$. All lines have an absorptive shape, with the intensity of the kth sideband proportional to $\mathrm{J}_k^2(z)$. The first-harmonic spectrum consists of in-phase components with absorptive lineshapes (Fig. 20c) , and out-of-phase components with dispersive lineshapes (Fig. 20d). The spectra are anti-symmetric with respect to the center line. The second-harmonic spectra shown in Figs. 20e,f are again symmetric.

In cw EPR spectroscopy the first-harmonic absorption signal in phase with the modulation frequency is usually measured (Fig. 20c). The derivative-like lineshape is then a superposition of a large number of overlapping sidebands separated by ω_{rf}. The two multiple-photon transitions with $|k|$ of approximately $2\omega_2/\omega_{\mathrm{rf}}$ give the maximum contributions to the signal.

For a homogeneous line and $2\omega_2 \ll 1/T_2$, Eq. (29) reduces to the commonly observed derivative lineshape

$$\langle S_y\rangle_1 \approx 2\omega_2\;\frac{\mathrm{d}}{\mathrm{d}\Omega_S}\left(\frac{\omega_1 T_2}{1+\Omega_S^2\;T_2^2}\right)\;. \tag{30}$$

The signal amplitude is proportional to the modulation amplitude, and does not depend on the modulation frequency.

Eq. (30) can be extended to the more general case of an inhomogeneous broadened line, which consists of a distribution of unresolved homogeneous lines. As long as the modulation amplitude does not exceed the inhomogeneous linewidth, again the derivative of the absorption line is observed.

It is noteworthy that in the first-harmonic in-phase cw EPR spectrum the observed derivative lines consist of a large number of multiple-photon transitions,

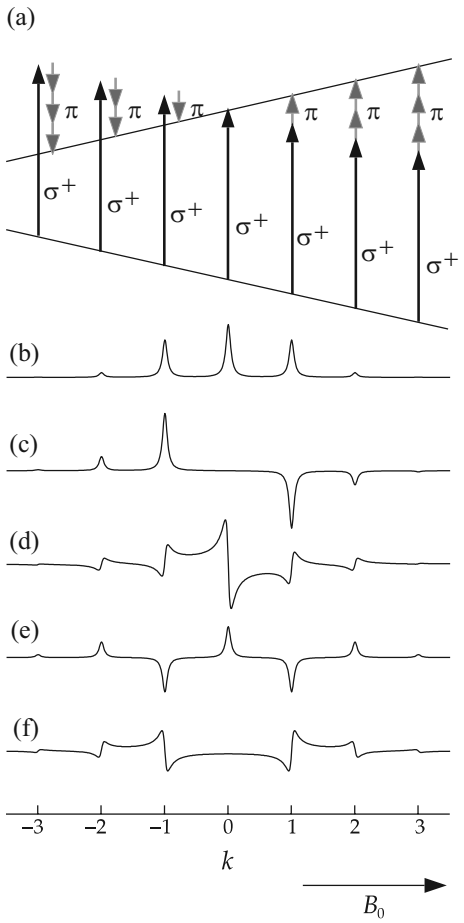


Fig. 20: Simulated field-swept cw EPR spectra for different harmonics of the modulation frequency. k indicates the number of absorbed rf π photons. (a) Multiple-photon transitions. (b) Zeroth-harmonic spectrum. (c) First-harmonic spectrum, in-phase with the radio frequency. (d) Corresponding out-of-phase spectrum. (e) Second-harmonic spectrum, in-phase with the radio frequency. (f) Corresponding out-of-phase spectrum (adapted from [72]).

and that the signal intensity is zero at the position of the single-photon transition. For example, in a standard cw EPR experiment with 100-kHz field modulation and a modulation amplitude of 0.1 mT ($2\omega_2/2\pi \approx 3$ MHz), the two multiple-photon transitions constituting the peak maximum and the peak minimum are $\sigma^+ \pm 28 \times \pi$ transitions, where 28 rf photons are absorbed and emitted.

The finding that the lineshapes observed in cw EPR experiments are caused by multiple-photon transitions was verified by experiments on a single crystal of lithium phthalocyanine with a linewidth of 4-8 μT. Two series of first-harmonic in-phase EPR spectra with different modulation amplitudes are shown in Fig. 21. The spectra recorded with a 1-MHz field modulation (Fig. 21a) show the patterns predicted by Eq. (29).

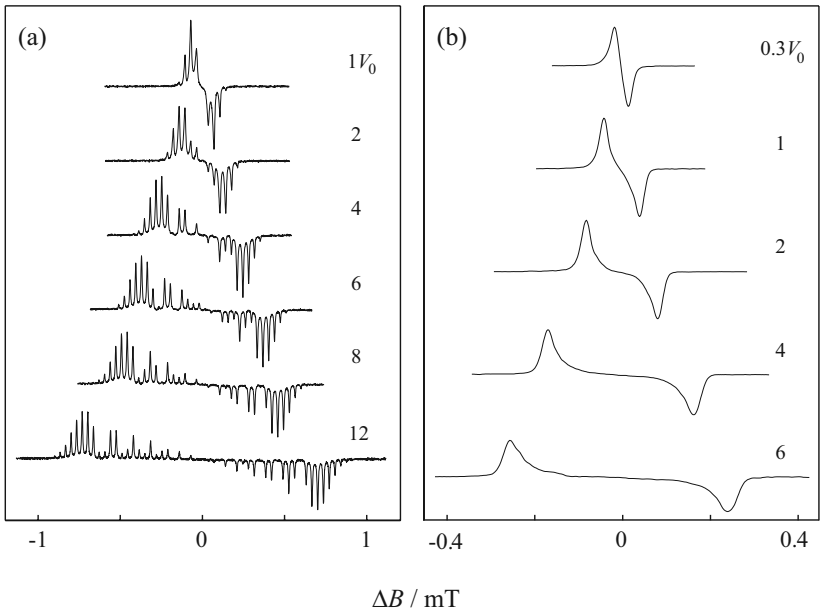


Fig. 21: First-harmonic EPR spectra of lithium phthalocyanine recorded at Q-band with different modulation amplitudes in phase with the modulation frequency. Numbers are proportional to the signal voltage. (a) Modulation frequency 1 MHz. (b) Modulation frequency 100 kHz (adapted from [72]).

When a 100-kHz field modulation is used (Fig. 21b) , the individual multiple-photon transitions are no longer resolved. The out-of-phase signal (not shown) is averaged to zero, since the individual multiple-photon transitions with derivative lineshape (see Fig. 20d) compensate each other.

Although it was not realized during many years since the beginning of EPR, in the traditional cw experiment we actually observe *multiple-photon transitions*. This recent finding gives a deeper insight into the origin of the observed phenomena and the interpretation of the results.

3.2 Pulse EPR experiments

We now demonstrate that multiple-photon transitions created by a bichromatic radiation field can also be observed in *pulse* EPR experiments. In particular, the creation of multiple-photon echoes is discussed.

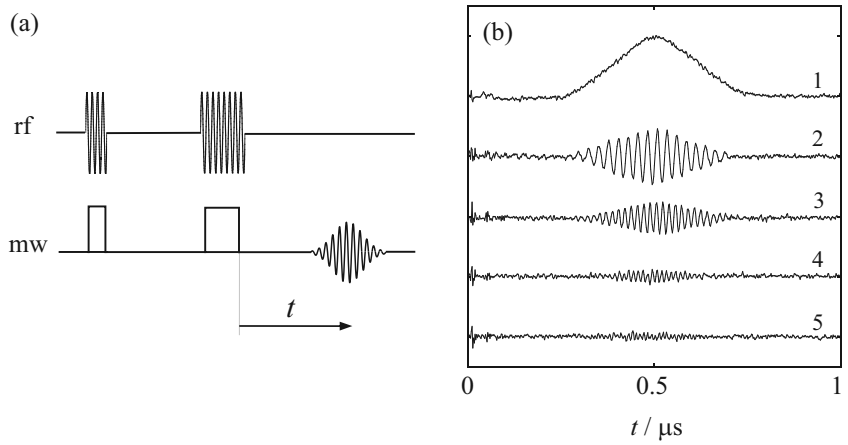


Fig. 22: Multiple-photon pulse EPR with bichromatic pulses. (a) Pulse sequence for a multiple-photon echo. (b) Observed time-domain echo signals. 1: Single-photon echo, $\omega_S = \omega_{\rm mw}$, $\omega_1/2\pi = 2.5$ MHz. 2-5: Two-photon echoes measured at $\Omega_S - \omega_{\rm mw} \approx \omega_{\rm rf}$, with $\Omega_S/2\pi$ (in MHz) 40 (2), 60 (3), 80 (4), and 100 (5) (adapted from [32]).

The pulse sequence for the generation of multiple-photon echoes is shown in Fig. 22a. The mw part consists of the conventional two-pulse sequence [18]. The sequence of rf pulses with the field along $\mathbf{B}_0$ is applied synchronously with the mw pulses. The first bichromatic pulse creates coherence at $\omega_{\rm mw}$ and $\omega_{\rm mw} + k\omega_{\rm rf}$. For simplicity we consider lowest (two-photon) transitions only $\omega_{\rm mw} \pm \omega_{\rm rf}$. During the free evolution period of time τ between the first and the second pulse the coherence dephases. The second bichromatic pulse inverts the directions of spin precession, resulting in the formation an echo at time τ after the second pulse. The echo appears again at $\omega_{\rm mw}$ and at $\omega_{\rm mw} \pm \omega_{\rm rf}$. By varying the mw and rf amplitude, as well as by choosing a proper $\mathbf{B}_0$ value, the echoes observed at different frequencies can be optimized or separated from each other, as is demonstrated in next paragraph [32, 73].

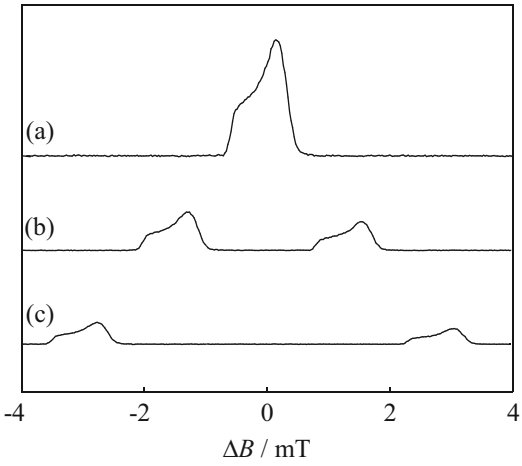


Fig. 23: Echo-detected single- and two-photon EPR spectra of γ-irradiated quartz glass. (a) Single-photon spectrum, $\omega_1/2\pi = 2.5$ MHz. (b) Two-photon spectrum with $\omega_{rf}/2\pi = 40$ MHz, $\omega_1/2\pi = \omega_2/2\pi = 10$ MHz. (c) Two-photon spectrum with $\omega_{rf}/2\pi = 80$ MHz, $\omega_1/2\pi = \omega_2/2\pi = 15$ MHz (adapted from [32]).

Two-photon echoes on γ-irradiated quartz glass are recorded at Q-band frequencies [32]. The lengths of the bichromatic pulses are 100 ns and 200 ns and the mw and rf amplitudes are both approximately 10 MHz. With these pulse sequences the signal of the conventional echo is maximum for an mw amplitude of about 2.5 MHz (Fig. 22b, trace 1). The echo traces 2–5 in Fig. 22b are obtained at B_0 fields corresponding to resonance offsets of 37.8, 58.2, 78.4 and 97.4 MHz, which correspond to the radio frequencies used in these experiments.

The echo-detected two-photon EPR spectra obtained by Fourier transformation of the time-domain traces and recorded with radio frequencies of 40 and 80 MHz are shown in Fig. 23b,c. As a reference, the single-photon EPR spectrum is measured with the same sequence. The two-photon EPR spectra are symmetrically placed to the position of the single-photon EPR spectrum, with shapes that are exact replicates of the latter. The observed peak-to-peak splittings of the two-photon EPR spectra are approximately twice the radio frequency.

3.3 π-photon-induced transparency

In this section we describe a new transparency phenomenon which can be observed in spin systems. For a two-level system exposed to a bichromatic radiation field *resonant* with the mw frequency, $\omega_{mw} = \omega_S$, energy and angular momentum are not only conserved for the single-photon transition but also for all multiple-photon transitions of the type $-m \times \pi + \sigma^+ + m \times \pi$, with $1 < m < \infty$ (Fig. 24a).

In the single-photon process (1), one mw σ^+ photon is absorbed, whereas in the two three-photon processes (2) with $m = 1$, one mw σ^+ photon is absorbed,

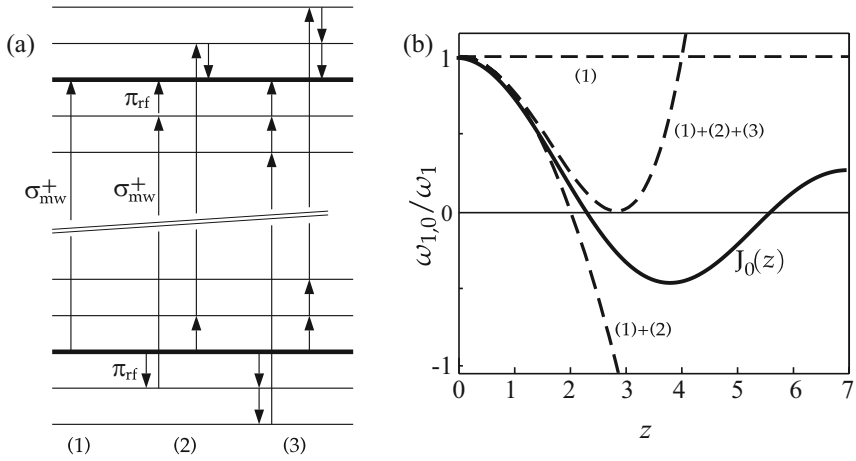


Fig. 24: (a) Multiple-photon transitions of the type $-m \times \pi + \sigma^+ + m \times \pi$. (1) Single-photon process σ^+; (2) three-photon processes, $\pi + \sigma^+ - \pi$ and $-\pi + \sigma^+ + \pi$; (3) five-photon processes $2\pi + \sigma^+ - 2\pi$ and $-2\pi + \sigma^+ + 2\pi$. (b) Normalized effective field $\omega_{1,0} = \omega_1 J_0(z)$ for the centre band at $\omega_{mw} = \omega_S$, as a function of $z = 2\omega_2/\omega_{rf}$ (bold solid line). The contributions of the processes (1), (1)+(2) and (1)+(2)+(3) are also given (dashed lines). Adapted from [73].

and one rf π photon is absorbed and one is emitted. Also five-photon transitions (3) with $m = 2$ and transitions of higher order are induced.

For properly chosen experimental parameters of the rf field, the spin system becomes transparent owing to destructive interference of the single-photon and the multiple-photon processes [76]. Since this phenomenon is caused by rf π photons, we call it *π-photon-induced transparency*.

For the description of this type of multiple-photon transitions the phases φ_{mw} and φ_{rf} of the mw and rf field have to be taken into account. Starting from the laboratory frame Hamiltonian

$$\mathcal{H}_{lab}(t) = \omega_S S_z + 2\omega_1 \cos(\omega_{mw} t + \varphi_{mw}) S_x + 2\omega_2 \cos(\omega_{rf} t + \varphi_{rf}) S_z \,, \qquad (31)$$

one arrives at the effective spin Hamiltonian in the toggling frame

$$\mathcal{H}_{TF,k} = ([\omega_S + \Delta_k] - [\omega_{mw} + k\,\omega_{rf}])\, S_z + \omega_{1,k}\, e^{-i\varphi_k S_z} S_x\, e^{i\varphi_k S_z} \,, \qquad (32)$$

where $\varphi_k = \varphi_{mw} + k\varphi_{rf}$ is the phase of the effective field. The Hamiltonian in Eq. (32) essentially describes the spin system exposed to an effective radiation field with frequency $\omega_{mw} + k\omega_{rf}$, phase φ_k, and amplitude $\omega_{1,k}$.

The effective transition amplitudes for the single-photon transition and the contributions of the processes (1), (1)+(2) and (1)+(2)+(3) are plotted in Fig. 24b as a function of the normalized rf amplitude $z = 2\omega_2/\omega_{rf}$. The effective transition

amplitude of the sum of *all* multiple-photon processes is then given by

$$\omega_{1,0} = \omega_1 \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{z}{2}\right)^{2m}}{(m!)^2} = \omega_1 \, \mathrm{J}_0(z) \ . \tag{33}$$

For the flip angle of a bichromatic pulse we find

$$\beta_{\mathrm{eff}} = \omega_{1,0} \, t_p = \omega_1 \, \mathrm{J}_0(z) \, t_p \ . \tag{34}$$

Corresponding expressions $\omega_{1,\mathrm{k}}$ can be derived for each of the sidebands.

Of special interest is the finding that the effective transition amplitudes $\omega_{1,k}$ can be *zero*; the two-level system then becomes transparent. For the center band with $\omega_S = \omega_{\mathrm{mw}}$, the effective transition amplitude $\omega_{1,0}$ is zero at the zero-crossings of the Bessel function $J_0(z)$. The first zero-crossing is at $z = j_{0,1} \approx 2.4048$, the second one at $z = j_{0,2} \approx 5.5201$, etc.

Fig. 25 describes the motion of the tip of the magnetization vector **M** in the rotating frame during bichromatic radiation with an mw field of amplitude ω_1 along the *x*-axis and an rf field $2\omega_2 \sin(\omega_{\mathrm{rf}} t)$, with $\omega_{\mathrm{rf}}/2\pi = 10$ MHz, along the *z*-axis.

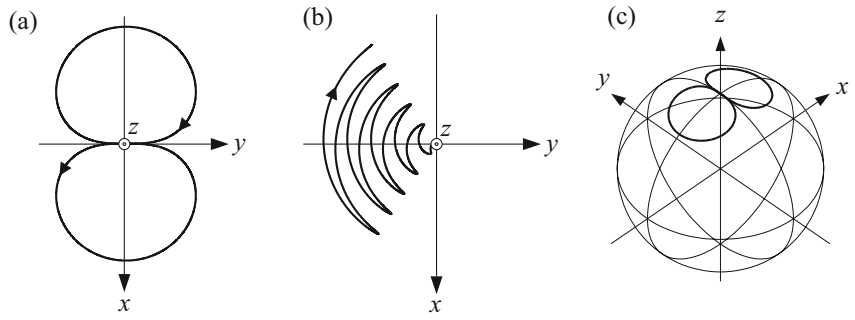


Fig. 25: Trajectories of the magnetization vector **M** under a bichromatic field, starting from thermal equilibrium (**M** along the *z*-axis). (a) Weak mw field, $\omega_1/2\pi$ = 0.01 MHz, resonant with the single-photon transition. Transparency condition fulfilled, $z = 2.4048$. Projection onto the *xy*-plane. (b) Transparency condition not fulfilled, $z = 1$. The trajectory is shown for the first five rf periods. The scaling of the axes is reduced by the factor 20 compared to (a). (c) Strong on-resonant mw field, $\omega_1/2\pi$ = 5 MHz. The transparency condition $z = 2.3306$ is fulfilled (adapted from [73]).

The situation for a weak mw field with amplitude $\omega_1/2\pi$ = 0.01 MHz, typically used in cw EPR experiments and $z = 2.4$ (first transparency condition) is shown in Fig. 25a. When the radiation field is turned on ($t = 0$), the effective field vector is given by $\boldsymbol{\omega}_{\mathrm{eff}} = \boldsymbol{\omega}_1$ and **M**, which is oriented along the *z*-axis, starts to nutate around *x*. Since $\omega_2 \gg \omega_1$, the angle between the effective field $\boldsymbol{\omega}_{\mathrm{eff}}$ and the *z*-axis gets rapidly very small, so that $\boldsymbol{\omega}_{\mathrm{eff}} \approx 2\boldsymbol{\omega}_2$. After one quarter of the rf period the magnetization is in the *xz*-plane, with a deviation from the *z*-axis of only 0.07°.

At half of the rf period ($t = \pi/\omega_{rf}$), **M** has moved back to the *z*-axis and the effective field is again $\boldsymbol{\omega}_{eff} = \boldsymbol{\omega}_1$. During the second half of the rf period, **M** describes a corresponding trajectory with negative *x*-values, so that after the full rf period **M** is again oriented along the *z*-axis. Thus, the tip of the magnetization vector **M** describes a *figure of eight* in the very close vicinity of the *z*-axis. If the transparency condition is not fulfilled, the trajectory is no longer a closed curve, and the magnetization moves on a toggling path towards the *xy*-plane, as is shown in Fig. 25b for the first five rf periods and $z = 1$.

For strong mw fields, as used in pulse EPR experiments, the third-order contribution $c_0^{(3)}$ (Eq. (24)) to the effective field amplitude can no longer be neglected. This reduces the value of *z* for which transparency is observed. For example, for an mw field with amplitude $\omega_1/2\pi$ = 5 MHz, corresponding to a π pulse with a length of 100 ns, transparency occurs at $z = 2.3306$ instead of $z = 2.4048$. The tip of **M** describes again a figure of eight, but with larger deviations from the *z*-axis (Fig. 25c).

Bichromatic pulses can be used as a tool to experimentally control the transition amplitude. This also includes the full suppression of the interaction of the mw field with the spin system. For example, during a resonant mw pulse a free evolution period can *artificially* be created by an rf pulse with a field amplitude that fulfills the first transparency condition. In this way an electron spin echo can be observed using a *single* mw pulse [76].

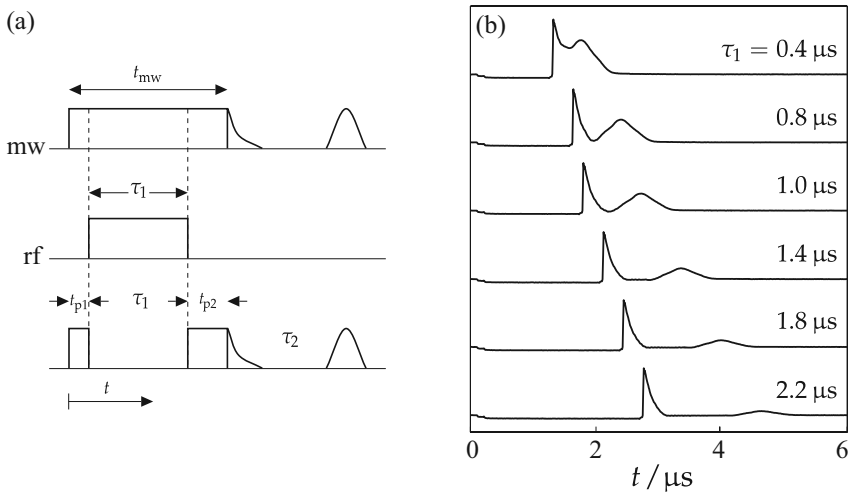


Fig. 26: One-pulse echo experiment with a π-photon-induced free evolution period on a coal sample, $\omega_{mw}/2\pi = 9.626$ GHz, $\omega_{rf}/2\pi = 15$ MHz, B_0 =341.1 mT. (a) Pulse sequence with $t_{p1} = 300$ ns, $t_{p2} = 600$ ns, $t_{mw} = 1.3 - 3.1$ µs, $\tau_1 = 0.4 - 2.2$ µs. The rf pulse applied during time τ_1 fulfills the first transparency condition for the single-photon transition. (b) Experimental time traces showing FIDs and echoes for different τ_1 values (adapted from [73]).

The pulse sequence for such a one-pulse echo experiment is shown in Fig. 26a. During the mw pulse of length t_{mw}, the rf pulse of length τ_1 is turned on at time t_{p1} and turned off at time $t_{mw} - t_{p2}$, with $2t_{p1} = t_{p2}$ and $\tau_1 = t_{mw} - t_{p1} - t_{p2}$. The mw field amplitude is chosen such that the nominal flip angle is $\beta_1 = \pi/2$ during t_{p1} and $\beta_2 = \pi$ during t_{p2}.

The pulse scheme thus corresponds to a two-pulse echo sequence, $(\pi/2) - \tau_1 - (\pi) - \tau_2 -$ (echo), with a *π-photon-induced free evolution period* of time τ_1, and a usual free evolution period of time $\tau_2 = \tau_1$. During time τ_1 the spin coherence evolves as it would do during a free evolution period, apart from a phase shift that depends on z, the rf phase, and the length of the bichromatic pulse.

Experimental time traces of such one-pulse echo experiments are shown in Figure 26b for different τ_1 values. The signal that follows the mw pulse consists of an FID and an electron spin echo at time $\tau_2 = \tau_1$, as in a conventional two-pulse echo experiment.

The bichromatic pulse as a substitute for a second mw frequency

The π-photon-induced transparency phenomenon can be applied, for example, in pulse EPR experiments that require two mw frequencies $\omega_{mw,1}$ and $\omega_{mw,2}$. A bichromatic pulse, which is transparent for the mw frequency $\omega_{mw} = \omega_{mw,1}$, may then be used as a substitute for the second mw frequency $\omega_{mw,2}$ [77].

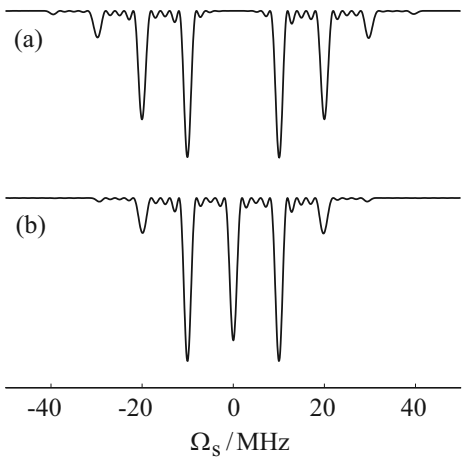


Fig. 27: Simulation of the polarization patterns of an $S = 1/2$ spin system with an inhomogeneous linewidth $\Gamma_{inh} = \infty$ after a bichromatic pulse with a length of 500 ns and a radio frequency of 10 MHz, as a function of Ω_S. (a) Transparency condition fulfilled, $z = 2.4$. The intensity of the center band is zero. (b) Transparency condition not fulfilled, $z = 1.5$ (adapted from [77]).

When the transparency condition is fulfilled for the center band at ω_{mw}, the bichromatic pulse only drives efficiently two- and three-photon transitions with resonance frequencies $\omega_{\mathrm{mw}} \pm \omega_{\mathrm{rf}}$ and $\omega_{\mathrm{mw}} \pm 2\omega_{\mathrm{rf}}$.

This is illustrated in Fig. 27, which shows simulations of the M_z magnetization of an $S = 1/2$ spin system with an inhomogeneous linewidth $\Gamma_{\mathrm{inh}} = \infty$ after a bichromatic π pulse of length 500 ns, as a function of the resonance off-set Ω_S. If the transparency condition is fulfilled (z=2.4), only spins with resonance frequencies $\omega_{\mathrm{mw}} \pm \omega_{\mathrm{rf}}$, $\omega_{\mathrm{mw}} \pm 2\omega_{\mathrm{rf}}$, and $\omega_{\mathrm{mw}} \pm 3\omega_{\mathrm{rf}}$ are excited (Fig. 27a). The contribution of higher-order multiple-photon transitions can be neglected.

A bichromatic pulse of proper length, fulfilling the transparency condition, can thus efficiently be used for broadband excitation with frequencies different from the mw frequency. If the transparency condition is not fulfilled, transitions at ω_{mw} and $\omega_{\mathrm{mw}} + k\omega_{\mathrm{rf}}$ are simultaneously excited. This is shown in Fig. 27b for z=1.5. For the transverse magnetizations M_x and M_y an analogous behavior is found, reflecting the excitation at frequencies $\omega_{\mathrm{mw}} + k\omega_{\mathrm{rf}}$ by the bichromatic pulse.

Stimulated Soft ESEEM

Stimulated Soft Electron Spin Echo Envelope Modulation (SS-ESEEM) is a pulse EPR experiment that is used to determine weak hyperfine and nuclear quadrupole interactions in paramagnetic species in solids [78]. The pulse sequence of SS-ESEEM shown in Fig. 28a makes use of two mw frequencies.

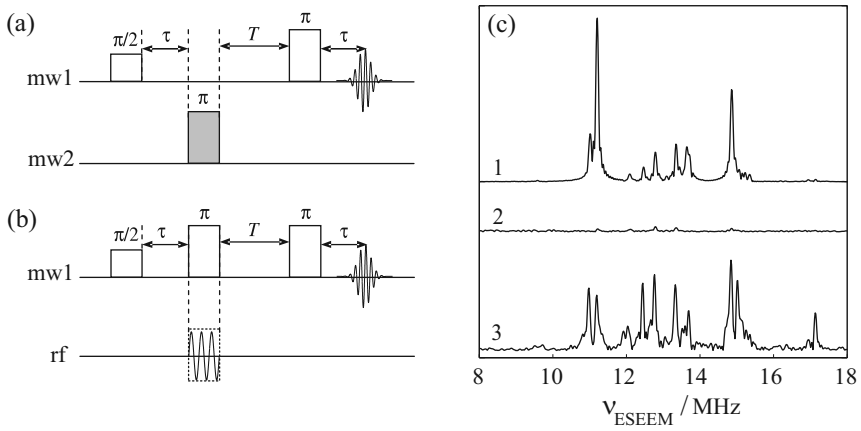


Fig. 28: Stimulated soft ESEEM. (a,b) Pulse sequences for SS-ESEEM (a), and bichromatic-pulse SS-ESEEM (b). (c) Comparison of three-pulse ESEEM and bichromatic-pulse SS-ESEEM experiments on a single crystal of ^{63}Cu(II)-doped Zn(picolinate)$_2$, arbitrary orientation, T=20 K. 1: Three-pulse ESEEM spectrum obtained with mw pulses with a length of 20 ns. 2: Corresponding spectrum obtained with mw pulses with a length of 400 ns. 3: Bichromatic-pulse SS-ESEEM spectra obtained with pulses with a length of 400 ns and a radio frequency of 14 MHz. Plots (2) and (3) use the same scale (adapted from [77]).

The first pulse with flip angle $\pi/2$ and frequency $\omega_{\mathrm{mw},1}$ creates electron coherence on an allowed (forbidden) transition. The second pulse with flip angle π and frequency $\omega_{\mathrm{mw},2}$ is resonant with a forbidden (allowed) transition and transfers the electron coherence to nuclear coherence, which then evolves during the following free evolution period of time T. The third pulse with flip angle π and frequency $\omega_{\mathrm{mw},1}$ transfers the nuclear coherence back to forbidden (allowed) electron coherence, which refocuses to a coherence-transfer echo. In this experiment, time T is incremented, and the modulation of the echo amplitude is monitored. SS-ESEEM experiments provide the same information as three-pulse ESEEM. They have, however, the advantages to be free of blind spots, and to require an mw power which is about two orders of magnitudes lower than for the standard three-pulse ESEEM experiment.

In the bichromatic version of SS-ESEEM, the second pulse with frequency $\omega_{\mathrm{mw},2}$ is replaced by a bichromatic pulse (Fig. 28b) [77]. The general features of SS-ESEEM are discussed elsewhere [78], here we concentrate on the characteristics of the experiment with a bichromatic pulse.

The potential of bichromatic-pulse SS-ESEEM is demonstrated on a single crystal of ^{63}Cu(II)-doped Zn(picolinate)$_2$. All the spectra are measured at a temperature of 20 K with an arbitrary crystal orientation. The three-pulse ESEEM spectrum shown in Fig. 28c, trace 1, is obtained with short non-selective mw pulses with a length of 20 ns, and consists of several proton peaks close to the proton Larmor frequency of 12.7 MHz.

Trace 2 and 3, demonstrate the superiority of SS-ESEEM over three-pulse ESEEM, when long low-power $\pi/2$ and π pulses with a length of 400 ns are used. Under this condition, three-pulse ESEEM is no longer operative, but SS-ESEEM still is. Trace 2 shows the three-pulse ESEEM spectrum. The excitation bandwidth of a pulse with a length of 400 ns is by far too small to simultaneously excite allowed and forbidden transitions. Trace 3 shows the bichromatic-pulse SS-ESEEM spectrum measured with two mw pulses and a bichromatic pulse with a radio frequency of 14 MHz. The improvement in the signal/noise ratio is tremendous.

The line positions in three-pulse ESEEM and the bichromatic-pulse SS-ESEEM spectrum are the same. Since SS-ESEEM does not suffer from blind spots, some of the lines are more intense in the bichromatic-pulse experiment than in three-pulse ESEEM. The line intensities are different in the two spectra, because the selective bichromatic pulse excites a narrow band of frequencies close to $\omega_{\mathrm{mw}} + k\omega_{\mathrm{rf}}$. Bichromatic-pulse SS-ESEEM is easy to implement and does not require a precise tuning of the bichromatic pulse to the transparency condition.

Double electron-electron resonance

Another experiment that makes use of two mw frequencies is double electron-electron resonance (DEER) [79], which has recently been extended to four pulses [80]. DEER may be used for the determination of distances between two spins A and B in solids by measuring the dipole coupling between two unpaired electrons.

The pulse sequence for four-pulse DEER is shown in Fig. 29a. It consists of three pulses with frequency $\omega_{\mathrm{mw},1}$ and one pulse with frequency $\omega_{\mathrm{mw},2}$. The first pulse with flip angle $\pi/2$ and frequency $\omega_{\mathrm{mw},1}$ creates electron coherence of the

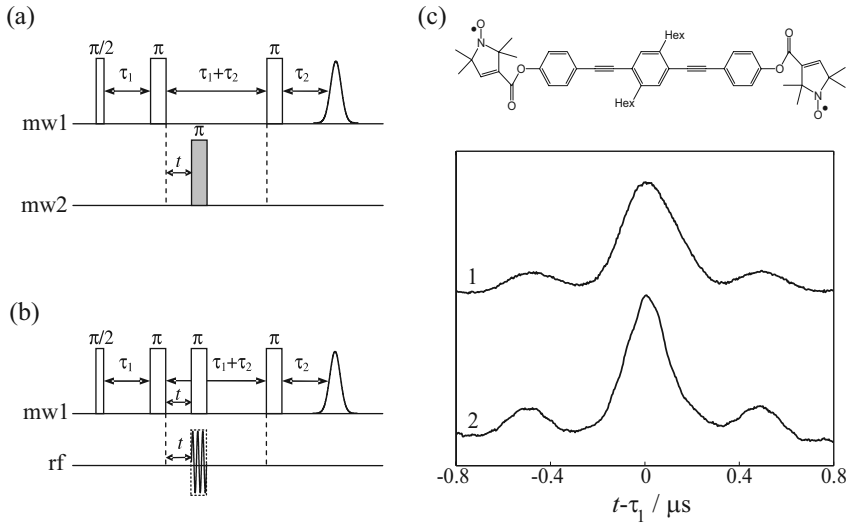


Fig. 29: Four-pulse DEER. (a,b) Pulse sequence for DEER (a), and bichromatic-pulse DEER (b). (c) Comparison of DEER time-traces of the nitroxide biradical CAS 312624-83-21 shown on top. 1: Standard four-pulse DEER with two mw frequencies, $(\omega_{mw,1} - \omega_{mw,2})/2\pi$=24 MHz. 2: Four-pulse DEER with a bichromatic pulse, $\omega_{rf}/2\pi$=24 MHz and $z \approx 2.4$ (adapted from [77]).

A-spins. The second pulse with flip angle π and frequency $\omega_{mw,1}$ refocuses this coherence to an electron spin echo, which is formed at time τ_1 after the pulse, and the third pulse with flip angle π and frequency $\omega_{mw,1}$ refocuses the coherence to a refocused echo. At time t after the second pulse an mw pulse with flip angle π and frequency $\omega_{mw,2}$ is applied to the B-spins. The effect of this pulse on the refocused echo intensity is recorded by varying time t between 0 and $2\tau_1$ ($\tau_1 < \tau_2$). The pulse with frequency $\omega_{mw,2}$ can again be replaced by a bichromatic pulse with mw frequency $\omega_{mw} = \omega_{mw,1}$ and radio frequency ω_{rf} (Fig. 29b).

Fig. 29c shows experimental DEER time traces obtained with the two-frequency experiment (trace 1) and with bichromatic-pulse DEER (trace 2). The time traces are very similar, and the dipolar coupling obtained from the corresponding spectra agrees well with results reported earlier [80].

In the case of DEER, it is important that the bichromatic pulse fulfills the transparency condition. Otherwise the residual excitation of A-spins at frequency ω_{mw} destroys the relevant signal, since A-spins and B-spins with resonance frequencies ω_{mw} and $\omega_{mw}+k\omega_{rf}$ are both flipped simultaneously. A bichromatic pulse of at least one period of the radio frequency has to be used in bichromatic-pulse DEER. This leads to a loss in signal intensity compared to standard four-pulse DEER. On the other hand, in bichromatic-pulse DEER a larger number of spins with frequencies $\omega_{mw}+k\omega_{rf}$ are simultaneously excited, which again improves the

signal intensity.

The examples of SS-ESEEM and DEER demonstrate that bichromatic pulses can successfully be used to substitute pulses with a second mw frequency.

In this Chapter we described the different types of multiple-photon transitions that can be observed and used in EPR, from both a theoretical and an experimental point of view. We especially emphasized on the multiple-photon transitions induced by an additional longitudinal oscillating field. These transitions play a major role in traditional field-modulated cw EPR and are responsible for its derivative-like lineshapes. Moreover, π-photon-induced transparency based on this type of transitions has proved to be a very interesting and useful phenomenon. One area of applications refers to two-frequency pulse EPR experiments such as DEER, Soft ESEEM and others. In the future, we will study the multiple-photon phenomena in more detail and screen their potential for practical applications.

A Second quantization

A.1 Hamiltonian of the photon field

For a fully quantum-mechanical description of the interaction between matter and an electro-magnetic radiation field the oscillating magnetic fields have to be replaced by a set of quantized harmonic oscillators [3, 81], this is called second quantization. The Hamiltonian of a spin system in a radiation field is then given by

$$\mathcal{H} = \mathcal{H}_\mathrm{S} + \mathcal{H}_\mathrm{R} + \mathcal{H}_\mathrm{i} \ , \tag{35}$$

where $\mathcal{H}_\mathrm{S}$ is the Hamiltonian of the undisturbed system, $\mathcal{H}_\mathrm{R}$ describes the radiation field, and $\mathcal{H}_\mathrm{i}$ describes the magnetic dipole coupling between spin system and radiation field.

The radiation field is represented by a superposition of an infinite number of transverse electro-magnetic waves, propagating in all possible directions. The different modes, distinguished by their propagation vectors $\mathbf{k}$ with $|\mathbf{k}| = \omega_k c$, are orthogonal and do not interact with each other. Every mode has two independent orthogonal polarizations λ. The polarization of the magnetic field component of a mode is given by the vector $\beta_{\mathbf{k}\lambda}$. For a circular field it is more convenient to describe the polarization with the unit vectors $\boldsymbol{\beta}_{+1} = -(\boldsymbol{\beta}_x + \mathrm{i}\boldsymbol{\beta}_y)/\sqrt{2}$ (right circular polarization), $\boldsymbol{\beta}_{-1} = (\boldsymbol{\beta}_x - \mathrm{i}\boldsymbol{\beta}_y)/\sqrt{2}$ (left circular polarization) and $\boldsymbol{\beta}_0 = \boldsymbol{\beta}_z$ (linear polarization), instead of using the basis system of Cartesian unit vectors, $\{\boldsymbol{\beta}_x, \boldsymbol{\beta}_y, \boldsymbol{\beta}_z\}$.

Using the correspondence principle the Hamiltonian of the radiation field is found to be

$$\mathcal{H}_\mathrm{R} = \sum_{\mathbf{k}\lambda} \omega_k \left(a^\dagger_{\mathbf{k}\lambda} a_{\mathbf{k}\lambda} + \frac{1}{2}\mathbb{1} \right) \ , \tag{36}$$

with the photon *annihilation* and *creation* operators a and $a^\dagger$. The combined operator $N_{\mathbf{k}\lambda} = a^\dagger_{\mathbf{k}\lambda} a_{\mathbf{k}\lambda}$ is called the *photon number* operator, since it gives the number $n_{\mathbf{k}\lambda}$ of photon quanta ω_k present in an eigenstate $|n_{\mathbf{k}\lambda}\rangle_{\mathbf{k}\lambda}$ of mode $\mathbf{k}\lambda$,

$$N_{\mathbf{k}\lambda} |n_{\mathbf{k}\lambda}\rangle_{\mathbf{k}\lambda} = n_{\mathbf{k}\lambda} |n_{\mathbf{k}\lambda}\rangle_{\mathbf{k}\lambda} \ . \tag{37}$$

Consequently, the eigenstates of the operator $N_{\mathbf{k}\lambda}$ are called *number states*.

The polarization vector of a radiation mode does enter in the interaction Hamiltonian $\mathcal{H}_\mathrm{i}$, which describes the coupling between the magnetic field and the magnetic dipole of the spin,

$$\mathcal{H}_\mathrm{i} = -\frac{1}{\hbar} \mathbf{B}^\mathrm{T} \boldsymbol{\mu} = -\frac{1}{\hbar} \sum_{\mathbf{k}\lambda} \mathbf{B}^\mathrm{T}_{\mathbf{k}\lambda} \boldsymbol{\mu} \ . \tag{38}$$

The operator of the quantized magnetic field is given by

$$\mathbf{B}(\mathbf{r}) \quad = \quad \mathrm{i} \sum_{\mathbf{k}\lambda} \boldsymbol{\beta}_{\mathbf{k}\lambda} \sqrt{\frac{\hbar\, \omega_k\, \mu_0}{2V}} \left[a_{\mathbf{k}\lambda} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} - a^\dagger_{\mathbf{k}\lambda} \mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}} \right] \ . \tag{39}$$

For low frequencies and correspondingly long wavelengths, as used in NMR and EPR spectroscopy, one finds $\mathbf{k}^\mathrm{T}\mathbf{r} \ll 1$, so that in good approximation $\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{r}) = 1$,

over all positions **r** within the sample volume. In this *dipolar approximation* the interaction between the radiation field and the magnetic dipole is reduced to the energy of a dipole in a dipolar field. The interaction Hamiltonian of a certain mode $\mathbf{k}\lambda$ can then be written as

$$\mathcal{H}_\mathrm{i} = -\gamma\sqrt{\frac{\hbar\,\omega_k\,\mu_0}{2V}}\,\mathrm{i}\,\boldsymbol{\beta}_{\mathbf{k}\lambda}^\mathrm{T}\left(a_{\mathbf{k}\lambda}+a_{\mathbf{k}\lambda}^\dagger\right)\mathbf{S}\;, \tag{40}$$

with the square root describing the magnetic field produced by a single photon with frequency ω_k in a cavity of volume V. For a *linear* field mode oriented parallel to the j-axis ($j = x, y, z$) the interaction Hamiltonian is given by

$$\mathcal{H}_\mathrm{i} = \lambda\left[a_{\mathbf{k}\lambda}+a_{\mathbf{k}\lambda}^\dagger\right]S_j\;, \tag{41}$$

with an effective coupling factor λ. For a right circular field mode, $\boldsymbol{\beta}_{\mathbf{k}\lambda} = \mathrm{i}\boldsymbol{\beta}_{+1} = (\mathrm{i}, -1, 0)/\sqrt{2}$, we find for the corresponding Hamiltonian

$$\mathcal{H}_\mathrm{i} = \lambda\left[a_{\mathbf{k}\lambda}S^+ + a_{\mathbf{k}\lambda}^\dagger S^-\right]\;. \tag{42}$$

Since these coupling elements connect spin states with $\Delta m = 1$, a right circular field consists of photons with helicity m_J=1, i.e. σ^+-photons.

A.2 Coherent states

The quantum-electrodynamical equivalents to classical radiation fields are *coherent states* [2,82,83]. A coherent state $|\alpha\rangle$ is an eigenfunction of the annihilation operator, $a|\alpha\rangle = \alpha|\alpha\rangle$ with a complex eigenvalue α, leading to the definition

$$|\alpha\rangle = \sum_{n=0}^{\infty}\;\mathrm{e}^{-\frac{1}{2}|\alpha|^2}\frac{\alpha^n}{\sqrt{n!}}\;|n\rangle\;. \tag{43}$$

The populations of the different number states follow a Poisson distribution

$$|\langle n|\alpha\rangle|^2 = \mathrm{e}^{|\alpha|^2}\,\frac{|\alpha|^{2n}}{n!} \tag{44}$$

with an average photon number $\langle N\rangle = |\alpha|^2$.

A typical *semiclassical* interaction Hamiltonian of a linear radiation field, oriented perpendicular to the static magnetic field, is

$$\mathcal{H}_\mathrm{i,semicl} = 2\omega_1\cos(\omega t)S_x\;. \tag{45}$$

By definition it is equal to the corresponding quantized interaction Hamiltonian in Eq. (41), $\mathcal{H}_\mathrm{i,quant} = \mathcal{H}_\mathrm{i,semicl}$, which leads to a coupling factor of

$$\lambda = \frac{\omega_1}{\langle N\rangle^{1/2}}\;. \tag{46}$$

The fully quantized Hamiltonian finally results in

$$\begin{aligned}\mathcal{H} &= \mathcal{H}_S + \mathcal{H}_\mathrm{R} + \mathcal{H}_\mathrm{i,quant} \\ &= \omega_S S_z + \omega\left(a^\dagger a + \frac{1}{2}\mathbb{1}\right) + \frac{\omega_1}{\langle N\rangle^{1/2}}\,(a^\dagger + a)S_x\ ,\end{aligned} \tag{47}$$

with a function space spanned by $|m_S, n\rangle = |m_S\rangle \otimes |n\rangle$. Physically, this can be understood as spin states being dressed by a certain number of photons. The resulting states are thus called *dressed states* [84].

For low frequencies and high power as used in magnetic resonance experiments $\langle N\rangle$ is extremely large, which allows further simplifications, as is shown in the following section.

A.3 Floquet theory

For very high average photon numbers $\langle N\rangle$ the Poisson distribution in Eq. (43) approximates a Gaussian distribution with very narrow relative linewidth. Almost exclusively states with $n \approx \langle N\rangle$ are populated. For these the coupling terms $\langle m_S, n|\mathcal{H}_\mathrm{i,quant}|m_S - 1, n + 1\rangle$ in Eq. (47) are in very good approximation equal to $\frac{1}{2}\omega_1$. The quantized field Hamiltonian can then be replaced by a Hamiltonian in the semiclassical Floquet space [15, 74, 85],

$$\mathcal{H}_\mathrm{F} \approx \mathcal{H} - \left(\langle N\rangle + \frac{1}{2}\right)\omega\,\mathbb{1}\ . \tag{48}$$

The advantages of the Floquet Hamiltonian are the reduced number of considered photons (only differences of photon numbers are of interest) and the simpler mathematical treatment due to the periodicity of eigenfunctions and eigenvalues.

Floquet theory was introduced in magnetic resonance by Shirley [15] to describe transition probabilities of nuclear spins in an atomic beam, subject to a linear rf field. Later it was applied to analyze saturation-transfer EPR spectra, by calculating the steady-state solution of a stochastic Liouville equation under the influence of radiation and modulation fields [86–89]. Nowadays Floquet theory is used, for example, for the description of multiple-photon processes in NMR [16] and EPR [17, 32, 46], and for explaining experimental observations in magic angle spinning (MAS) NMR [74, 85].

Mathematically, the Floquet formalism is based on a solution of linear differential equations with periodic coefficients [90]. A semiclassical periodically time-dependent Hamiltonian $\mathcal{H}(t) = \mathcal{H}_0 + \mathcal{H}_1(t)$ in the eigenbasis of $\mathcal{H}_0$, with fundamental frequency $\omega/2\pi$ and period $t_p = 2\pi/\omega$, is expanded to a Fourier series,

$$\langle p|\mathcal{H}(t)|q\rangle = \sum_{n=-\infty}^{\infty} \langle p|h^n|q\rangle \mathrm{e}^{\mathrm{i}n\omega t}\ , \tag{49}$$

where h^n is the nth Fourier component of $\mathcal{H}(t)$. A Floquet Hamiltonian is defined as a Hermitian matrix of infinite dimension in Floquet space

$$\langle p, n|\mathcal{H}_\mathrm{F}|q, m\rangle = \langle p|h^{n-m}|q\rangle + n\omega\delta_{nm}\delta_{pq}\ , \tag{50}$$

where $|p,n\rangle$ represents a spin function $|p\rangle$ dressed by n quanta with frequency ω. The evolution of the density operator in Floquet space is described in analogy to the Hilbert space by

$$\sigma_{\mathrm{F}}(t) = U_{\mathrm{F}}(t;t_0)\sigma_{\mathrm{F}}(t_0)U_{\mathrm{F}}^{-1}(t;t_0)\ , \tag{51}$$

with the propagator

$$U_{\mathrm{F}}(t;t_0) = \mathrm{e}^{-\mathrm{i}\mathcal{H}_{\mathrm{F}}(t-t_0)}\ . \tag{52}$$

For the calculation of the evolution of the spin system in Hilbert space either the density operator or the propagator has to be transformed back to Hilbert space.

For time-dependent Hamiltonians with more than one basic frequency the Floquet theory is expanded to the *many-mode Floquet theory* [91,92]. When transient effects are investigated the evolution of the density operator in Hilbert space can be calculated numerically [93].

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